



Time Series Analysis and Forecasting

Chapter 5b: Multivariate GARCH Models



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Learning Objectives

By the end of this chapter, you will be able to:

1. Explain why **multivariate volatility modeling** is necessary
2. Formulate and estimate **VEC**, **BEKK**, **CCC**, and **DCC** models
3. Interpret **dynamic correlations** and covariance decompositions
4. Apply multivariate GARCH for **portfolio optimization**, **hedging**, and **risk management**

Practical Skills

- ▣ Python implementation with `rmgarch` / `mgarch` packages
- ▣ Interpreting time-varying covariance matrices
- ▣ Dynamic hedge ratios and portfolio weights

Key takeaway: *You are not learning models — you are learning how to measure real portfolio risk.*

Outline

Foundations

- ▣ Motivation: From Univariate to Multivariate
- ▣ The Multivariate GARCH Framework
- ▣ VEC and Diagonal VEC Models
- ▣ The BEKK Model
- ▣ Constant Conditional Correlation (CCC)
- ▣ Dynamic Conditional Correlation (DCC)

Applications

- ▣ Other Multivariate GARCH Models
- ▣ Estimation and the Curse of Dimensionality
- ▣ Model Selection and Diagnostics
- ▣ Portfolio Optimization
- ▣ Dynamic Hedging
- ▣ Summary and Quiz

The Story of This Chapter

The Problem

- ▣ Correlations between assets **change over time** — and they spike during crises
- ▣ Ignoring this leads to wrong portfolios, bad hedges, and underestimated risk

The Difficulty

- ▣ Modeling N assets requires $N(N + 1)/2$ time-varying covariance elements
- ▣ The **curse of dimensionality** makes most approaches infeasible

The Solutions (from general to practical)

- ▣ **VEC**: the most general model — but unusable in practice
- ▣ **BEKK**: guarantees positive definiteness — but does not scale
- ▣ **CCC**: simple and fast — but correlations are *constant* (wrong!)
- ▣ **DCC**: the optimal compromise — dynamic correlations with **only 2 extra parameters**

Why Go Multivariate?

The Problem

- Univariate GARCH models each asset's volatility **in isolation**
- They ignore **co-movements** between assets:
 - ▶ How does the volatility of asset *A* spill over to asset *B*?
 - ▶ How do correlations between assets change over time?

But in reality...

- Markets are interconnected — shocks propagate across assets
- Financial decisions require the **full covariance matrix** H_t , not just individual variances

Key insight: *Risk is not the sum of individual risks.*

The Central Question: Modeling H_t

Setup

- Given N assets with return vector $r_t = (r_{1t}, \dots, r_{Nt})'$, we need to model:
$$H_t = \text{Var}(r_t \mid \mathcal{F}_{t-1})$$

- the **time-varying conditional covariance matrix**

Why is this hard?

- H_t has $N(N+1)/2$ unique elements — for $N = 50$: **1,275 parameters** at each t
- Must be **positive definite** at every time step
- Must capture **dynamic** correlations, not just volatilities

Intuition: H_t is the “weather forecast” for risk — it updates daily as new information arrives.

Application 1: Portfolio Allocation

Markowitz with Time-Varying Inputs

- Optimal weights: $w_t^* = \frac{H_t^{-1} \mu_t}{1' H_t^{-1} \mu_t}$
- Weights shift **daily** as H_t updates
- Static covariance matrices lead to **suboptimal allocations** in turbulent periods

The Problem

- If H_t is treated as constant $\Rightarrow$ portfolio weights are wrong during crises
- Rebalancing too slowly $\Rightarrow$ excess risk; too fast $\Rightarrow$ excess cost

Key takeaway: Optimal portfolios are dynamic, not static.

Application 2: Dynamic Hedging

Minimum-Variance Hedge Ratio

- Optimal hedge: $h_t^* = \frac{\sigma_{12,t}}{\sigma_{22,t}}$
- Hedge more when spot-futures **covariance rises**
- Requires **time-varying covariances**, not just variances

Why Constant Hedging Fails

- In crisis: covariance structure shifts rapidly
- A fixed hedge ratio **under-hedges** or **over-hedges** depending on regime

***Key insight:** Protection depends on the relationship between assets, not on individual volatilities.*

Application 3: Risk Management (VaR)

Portfolio Value-at-Risk

- ▣ $\text{VaR}_t^p = -w' \mu_t + z_\alpha \sqrt{w' H_t w}$
- ▣ Risk depends on the **full covariance matrix**, not just individual volatilities

The Critical Problem

- ▣ Ignoring time-varying correlations $\Rightarrow$ **underestimates risk** during crises
- ▣ Correlations increase **precisely when diversification is needed most**

Key takeaway: Correlations rise exactly when you need diversification the most.

Stylized Fact: Correlations Are Not Constant

Empirical Evidence

- ▣ Correlations between assets are **not constant** — they increase during market turmoil
- ▣ This phenomenon is known as **correlation breakdown** or **contagion**
- ▣ Example: S&P 500 – DAX correlation jumped from 0.5 to 0.9 during the 2008 crisis
- ▣ **Diversification fails precisely when it is needed most**

***Key insight:** Correlations are regime-dependent — they spike during crises and compress during calm periods.*

Numerical Example: The Impact of Correlations on Risk

Balanced Portfolio: 50% Stocks, 50% Bonds

- ▣ $\sigma_1 = 20\%$ (stocks), $\sigma_2 = 5\%$ (bonds)
- ▣ Portfolio variance: $\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$

Regime	ρ	σ_p	VaR 99% (1 day)
Calm market	-0.20	9.22%	-2.14%
Normal market	+0.10	10.25%	-2.38%
Crisis (2008)	+0.60	11.73%	-2.73%

Interpretation: Portfolio risk increases by **27%** solely from the correlation change — it is the synchronization of movements, not individual volatility, that drives risk.

Why It Matters: Practical Implications

Constant-Correlation Models Are Dangerous

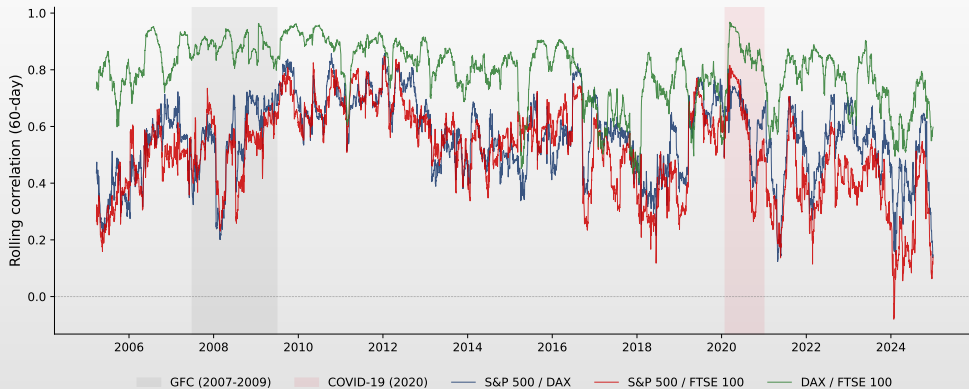
- ▣ **Portfolios:** weights based on stale correlations are suboptimal
- ▣ **Hedging:** fixed hedge ratios under-perform in regime shifts
- ▣ **VaR:** constant- ρ models underestimate tail risk during crises

Regulatory Implication

- ▣ Constant-correlation models are **inadequate** for:
 - ▶ Stress testing and scenario analysis
 - ▶ Real-time portfolio rebalancing
 - ▶ Regulatory capital calculations under Basel III/IV

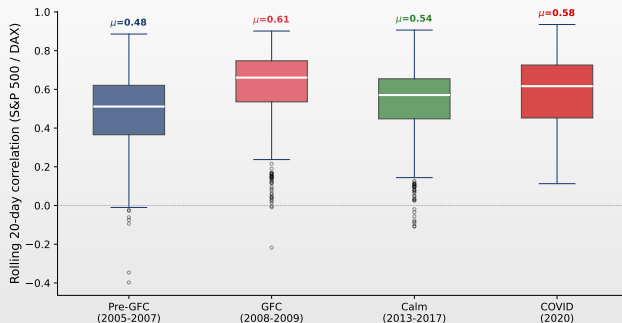
Key takeaway: Models with constant correlations are dangerous in practice.

Empirical Evidence: Rolling Correlations S&P 500, DAX, FTSE 100



TSA_ch5b_rolling_corr

Correlation Distribution: Crisis vs. Calm Regimes



Key Observations

- Median correlation increases significantly during crisis periods (GFC (*Global Financial Crisis*), COVID)
- Correlation dispersion rises — more uncertainty in cross-market relationships

Half-Vectorization — the $\text{vech}(\cdot)$ Operator

Definition

- The operator $\text{vech} : \mathbb{R}^{N \times N} \rightarrow \mathbb{R}^{N^*}$ stacks **column by column** the elements on and below the main diagonal of a symmetric matrix
- Output dimension: $N^* = N(N+1)/2$
- H_t is **symmetric**: $h_{ij,t} = h_{ji,t}$ — vech removes the $N(N-1)/2$ redundant elements

$$H_t = \begin{pmatrix} h_{11,t} & h_{12,t} & h_{13,t} \\ h_{12,t} & h_{22,t} & h_{23,t} \\ h_{13,t} & h_{23,t} & h_{33,t} \end{pmatrix} \xrightarrow{\text{vech}} \begin{pmatrix} h_{11,t} \\ h_{12,t} \\ h_{13,t} \\ h_{22,t} \\ h_{23,t} \\ h_{33,t} \end{pmatrix}$$

- Elements in gray are duplicates — vech removes them.

Central Role in Multivariate GARCH

- Allows writing the model as a vector equation: $\text{vech}(H_t) = c + A \text{vech}(\epsilon_{t-1} \epsilon'_{t-1}) + B \text{vech}(H_{t-1})$
- Dimension: $N^* = N(N+1)/2$ — for $N=2$: 3 elements; $N=10$: 55; $N=100$: 5,050

General Setup

Multivariate Return Process

- ▣ $\mathbf{r}_t = (r_{1t}, \dots, r_{Nt})'$: N -dimensional return vector

$$\mathbf{r}_t = \boldsymbol{\mu}_t + \boldsymbol{\varepsilon}_t \quad (1)$$

$$\boldsymbol{\varepsilon}_t = \mathbf{H}_t^{1/2} \mathbf{z}_t, \quad \mathbf{z}_t \stackrel{iid}{\sim} \mathcal{N}(0, \mathbf{I}_N) \quad (2)$$

- ▣ $\boldsymbol{\mu}_t$ = conditional mean vector (often a VAR or constant)
- ▣ $\mathbf{H}_t = \text{Var}(\boldsymbol{\varepsilon}_t \mid \mathcal{F}_{t-1}) = N \times N$ conditional covariance matrix
- ▣ $\mathbf{H}_t^{1/2}$ = any matrix square root such that $\mathbf{H}_t^{1/2}(\mathbf{H}_t^{1/2})' = \mathbf{H}_t$

Intuition: $\mathbf{H}_t^{1/2} \mathbf{z}_t$ transforms independent shocks into correlated returns — the “square root” imprints the dependence structure.

The Challenge

- ▣ $\mathbf{H}_t$ has $N(N+1)/2$ unique elements. For $N = 50$: **1,275 time-varying parameters** at each t !

Positive Definiteness Requirement

Why Positive Definiteness Matters

- ▣ The conditional covariance matrix H_t must be **positive definite** (p.d.) for all t :
- ▣ Portfolio variance: $\sigma_p^2 = w'H_t w > 0$ for all $w \neq 0$
- ▣ The log-likelihood involves $\log |H_t|$ and H_t^{-1} — both require p.d.
- ▣ Cholesky decomposition for simulation requires p.d.

Strategies to Ensure Positive Definiteness

1. **Restrict parameters** so that H_t is p.d. by construction (BEKK)
2. **Decompose** $H_t = D_t R_t D_t$ and model D_t , R_t separately (DCC)
3. **Work in factor space** to reduce dimensionality (Factor GARCH, GO-GARCH)

The Cholesky Decomposition

Definition

- Any **symmetric positive definite** matrix H ($N \times N$) admits a **unique factorization**: $H = LL'$
- L is a **lower triangular** matrix with **strictly positive** diagonal elements
- Complexity: $O(N^3/3)$ — **twice as fast** as direct inversion

Example: the 2×2 case

- Let $H = \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix}$. Then $L = \begin{pmatrix} \sqrt{h_{11}} & 0 \\ h_{12}/\sqrt{h_{11}} & \sqrt{h_{22} - h_{12}^2/h_{11}} \end{pmatrix}$

Role in Multivariate GARCH

- Simulation**: $\varepsilon_t = L_t z_t$, where $z_t \sim N(0, I)$ — generates correlated shocks
- Log-likelihood**: $\log |H_t| = 2 \sum_i \log L_{ii}$ — avoids direct determinant computation
- Inversion**: solve $L_t x = \varepsilon_t$ by forward substitution ($O(N^2)$ vs. $O(N^3)$)

Parameter Proliferation: The Curse of Dimensionality

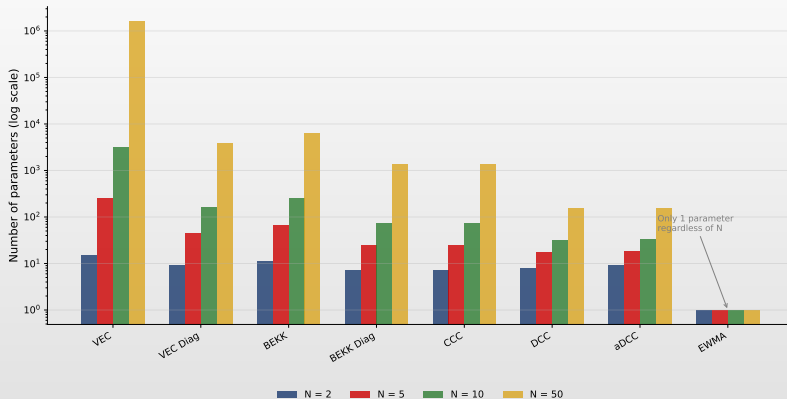
Number of Parameters by Model

Model	Parameters	$N = 5$
VEC(1,1)	$N(N+1)/2 + 2[N(N+1)/2]^2$	465
Diagonal VEC(1,1)	$3 \times N(N+1)/2$	45
BEKK(1,1)	$N(N+1)/2 + 2N^2$	65
Diagonal BEKK(1,1)	$N(N+1)/2 + 2N$	25
CCC(1,1)	$3N + N(N-1)/2$	25
DCC(1,1)	$3N + N(N-1)/2 + 2$	27

Practical Implication

- For large N , only **heavily parameterized models** (CCC, DCC, scalar BEKK, factor models) are feasible. Full VEC/BEKK become computationally intractable for $N > 5$.

Curse of Dimensionality: Visualization



TSA_ch5b_param_comparison

The Multivariate Normal Log-Likelihood

General Log-Likelihood

- For T observations of N -dimensional returns:

$$\mathcal{L}(\theta) = -\frac{TN}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \left[\underbrace{\log |H_t|}_{\text{volume penalty}} + \underbrace{\epsilon_t' H_t^{-1} \epsilon_t}_{\text{Mahalanobis distance}} \right]$$

Intuition: the likelihood penalizes both large covariance ("diffuse forecast") and large standardized residuals ("bad forecast").

Computational Challenges

- At each t : compute H_t ($O(N^2)$ or more), its inverse ($O(N^3)$), and determinant ($O(N^3)$)
- Cholesky trick:** if $H_t = L_t L_t'$, then $\log |H_t| = 2 \sum_i \log L_{ii}$ and solve $L_t x = \epsilon_t$ by forward substitution
- For DCC: two-step estimation dramatically reduces cost

The VEC Model — “Mother” of All MGARCH Models

: the most general linear MGARCH

- Every model you will learn (BEKK, CCC, DCC) is a **restricted version** of VEC
- The VEC(1,1) equation:

$$\text{vech}(H_t) = \underbrace{c}_{\text{intercept}} + \underbrace{A \text{vech}(\varepsilon_{t-1} \varepsilon'_{t-1})}_{\text{shock impact}} + \underbrace{B \text{vech}(H_{t-1})}_{\text{persistence}}$$

- $c: N^* \times 1$; $A, B: N^* \times N^*$; $N^* = N(N+1)/2$
- Full A allows every shock to affect every covariance element

Intuition: today's covariance = baseline + yesterday's shocks + yesterday's covariance.

Diagonal VEC: a Tractable Restriction

- $\text{vech}(H_t) = c + a \odot \text{vech}(\varepsilon_{t-1} \varepsilon'_{t-1}) + b \odot \text{vech}(H_{t-1})$ ($\odot$ = element-wise)
- Each variance/covariance follows its own GARCH — **no cross-effects**
- Reduces parameters from $O(N^4)$ to $3N^*$

VEC: Limitations and Why We Move On

► **Limitation:** VEC is practically unusable for $N > 2$

- ▣ For $N = 3$: A is $6 \times 6 \Rightarrow$ **78 parameters**; for $N = 5$: **465 parameters**
- ▣ **No guarantee** that H_t is positive definite
- ▣ Diagonal VEC reduces parameters but **kills all volatility spillovers**

► **When NOT to use VEC**

- ▣ **Never** use full VEC for $N > 2$ — estimation diverges
- ▣ **Avoid** Diagonal VEC for contagion studies — it cannot capture spillovers
- ▣ **Skip** VEC entirely if you need guaranteed positive definiteness

Model Positioning

- ▣ **Use when:** $N = 2$ and maximum flexibility is needed
- ▣ **vs. BEKK:** BEKK guarantees p.d. with same expressiveness — prefer BEKK

Key takeaway: VEC is the theoretical foundation — but in practice, we need something better.

From VEC to BEKK: Why We Need Guaranteed Positive Definiteness

What went wrong with VEC?

- VEC is the most **general** multivariate GARCH — every element of H_t has its own dynamics
- But: (1) **no guarantee** that H_t is positive definite, (2) parameter count **explodes** ($O(N^4)$)
- Diagonal VEC solves (2) but still fails at (1) — and loses spillover effects

The Core Problem

- VEC writes $\text{vech}(H_t) = c + A \text{vech}(\varepsilon_{t-1}\varepsilon'_{t-1}) + B \text{vech}(H_{t-1})$
- Nothing in this equation **forces** H_t to be p.d. at every t
- Sufficient conditions exist (Engle & Kroner, 1995) but are **hard to impose** during numerical optimization

BEKK's Insight: Build PD Into the Structure

- Write $H_t = CC' + A'\varepsilon_{t-1}\varepsilon'_{t-1}A + B'H_{t-1}B$
- Each term is p.s.d. **by construction** ($X'MX \succeq 0$ for any $M \succeq 0$)
- Sum of p.s.d. matrices + a p.d. matrix = p.d. always

The BEKK Model (Engle & Kroner, 1995)

Definition: BEKK(1,1)

$$H_t = \underbrace{CC'}_{\text{intercept (p.d.)}} + \underbrace{A'\varepsilon_{t-1}\varepsilon'_{t-1}A}_{\text{shock impact (p.s.d.)}} + \underbrace{B'H_{t-1}B}_{\text{persistence (p.s.d.)}}$$

- C : $N \times N$ lower triangular; A, B : $N \times N$ matrices

Intuition: the quadratic form $A'MA$ is always p.s.d. — positive definiteness comes “for free.”

Key Properties

- H_t is **positive definite by construction** (if CC' is p.d.)
- Captures both **own-volatility persistence** and **cross-volatility spillovers**
- Parameters: $N(N+1)/2 + 2N^2$ — for $N = 3$: $6 + 18 = 24$

: BEKK does not scale

- Conceptually important, but **infeasible for $N > 5$** (parameter explosion: $2N^2$)
- BEKK = **B**aba, **E**ngle, **K**raft, **K**roner (1990)

BEKK: Bivariate Case

Expanding the BEKK(1,1) for $N = 2$

□ Let $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, $C = \begin{pmatrix} c_{11} & 0 \\ c_{21} & c_{22} \end{pmatrix}$. Then:

$$h_{11,t} = c_{11}^2 + a_{11}^2 \varepsilon_{1,t-1}^2 + 2a_{11}a_{21}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{21}^2 \varepsilon_{2,t-1}^2 + b_{11}^2 h_{11,t-1} + 2b_{11}b_{21}h_{12,t-1} + b_{21}^2 h_{22,t-1} \quad (3)$$

$$h_{12,t} = c_{11}c_{21} + a_{11}a_{12}\varepsilon_{1,t-1}^2 + (a_{11}a_{22} + a_{12}a_{21})\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{21}a_{22}\varepsilon_{2,t-1}^2 + b_{11}b_{12}h_{11,t-1} + (b_{11}b_{22} + b_{12}b_{21})h_{12,t-1} + b_{21}b_{22}h_{22,t-1} \quad (4)$$

$$h_{22,t} = c_{21}^2 + c_{22}^2 + a_{12}^2 \varepsilon_{1,t-1}^2 + 2a_{12}a_{22}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{22}^2 \varepsilon_{2,t-1}^2 + b_{12}^2 h_{11,t-1} + 2b_{12}b_{22}h_{12,t-1} + b_{22}^2 h_{22,t-1} \quad (5)$$

Interpretation

□ Cross terms like $a_{21}^2 \varepsilon_{2,t-1}^2$ in $h_{11,t}$: shock in asset 2 affects asset 1's variance = **volatility spillover**.

Diagonal and Scalar BEKK

Diagonal BEKK

- Restrict A and B to be **diagonal**:

$$H_t = CC' + \text{diag}(a)' \varepsilon_{t-1} \varepsilon_{t-1}' \text{diag}(a) + \text{diag}(b)' H_{t-1} \text{diag}(b)$$

- Parameters: $N(N+1)/2 + 2N$. Element-wise: $h_{ij,t} = c_{ij}^* + a_i a_j \varepsilon_{i,t-1} \varepsilon_{j,t-1} + b_i b_j h_{ij,t-1}$

Scalar BEKK

- Further restrict: $A = aI_N$, $B = bI_N$:

$$H_t = CC' + a^2 \varepsilon_{t-1} \varepsilon_{t-1}' + b^2 H_{t-1}$$

- Parameters: $N(N+1)/2 + 2$. All pairs share the **same dynamics**.

Intuition: "one knob for shock sensitivity, one for memory" — identical across all pairs.

Trade-off

- Diagonal/Scalar BEKK: feasible for large N , but **no cross-asset spillovers**.

The Kronecker Product

Definition

▣ Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$. The **Kronecker product** $A \otimes B$ is the $mp \times nq$ block matrix:

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}$$

▣ Dimensions do **not** need to match (unlike Hadamard)

Example: 2×2

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \otimes \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix} \quad (\text{result is } 4 \times 4)$$

Role in Multivariate GARCH

- ▣ BEKK stationarity condition: $\rho(A \otimes A + B \otimes B) < 1$
- ▣ Unconditional covariance: $\text{vech}(\bar{H}) = (I - A \otimes A - B \otimes B)^{-1} \text{vech}(CC')$

BEKK: Stationarity and Unconditional Covariance

Stationarity Condition

- The BEKK(1,1) process is covariance stationary if and only if:

$$\rho(A \otimes A + B \otimes B) < 1$$

- where $\rho(\cdot)$ denotes the spectral radius and $\otimes$ is the Kronecker product

Intuition: the multivariate analogue of $\alpha + \beta < 1$ — shocks must decay, not explode.

Unconditional Covariance Matrix

- If the stationarity condition holds:

$$\text{vech}(\bar{H}) = (I_{N^*} - A \otimes A - B \otimes B)^{-1} \text{vech}(CC')$$

Intuition: the long-run covariance matrix that H_t gravitates toward — like $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$ in univariate GARCH.

BEKK: Strengths, Weaknesses, and Variant Selection

Strengths

- ▣ **Positive definiteness by construction** — no additional restrictions needed
- ▣ Captures **volatility spillovers** — the only class that directly models them
- ▣ Direct interpretation of A and B: shock transmission and persistence

Weaknesses

- ▣ **Parameter explosion**: full BEKK for $N = 10 \Rightarrow 255$ parameters
- ▣ Numerically **unstable** optimization — multiple local optima, slow convergence
- ▣ Results can be **sensitive to starting values**
- ▣ Cannot be decomposed into stages — joint estimation required

Practical Rule

- ▣ $N \leq 3$: full BEKK (if spillovers matter)
- ▣ $3 < N \leq 10$: diagonal BEKK
- ▣ $N > 10$: switch to DCC or factor models

BEKK: Key Insight and Common Pitfalls

: BEKK is a “smart reparametrization” of VEC

- BEKK does not add new economics — it writes the *same* dynamics in a p.d.-guaranteed form
- $A'MA$ is p.s.d. by linear algebra, not economic theory; trades flexibility for convenience

: using full BEKK for $N > 5$

- Full BEKK for $N = 10$: **255 parameters** — optimizer finds local optima, results are fragile
- Caporin & McAleer (2012): “full BEKK is essentially uninterpretable for $N > 3$ ”
- Diagonal BEKK is often the practical choice, but then spillovers vanish — defeating BEKK's purpose

- $N > 5$: use DCC or factor models instead
- When you don't care about *direction* of spillovers (DCC captures co-movement without spillover channels)
- When you need fast re-estimation (BEKK: hours vs. DCC: seconds for $N = 50$)

BEKK Section Summary

3 Things to Remember

1. BEKK guarantees p.d. via the quadratic form $A'MA$ — the **only** model that directly captures spillovers
2. Full BEKK: rich but infeasible for $N > 5$; diagonal BEKK: scalable but no spillovers
3. Estimation requires joint optimization — no two-step shortcut

Model Positioning

- ▣ **Use when:** $N \leq 5$ and volatility spillover *direction* matters
- ▣ **vs. VEC:** same flexibility, guaranteed p.d. **vs. DCC:** DCC scales better but cannot identify spillover direction

From BEKK to CCC: The Decomposition That Changed Everything

What BEKK taught us — and where it breaks down

- BEKK **solved** the PD problem and models volatility spillovers directly
- But: $N(N+1)/2 + 2N^2$ parameters $\Rightarrow$ estimation collapses for $N > 5$
- Joint estimation of all parameters — no decomposition into simpler sub-problems
- Practitioners abandoned BEKK for large portfolios by the late 1990s

Bollerslev's Insight (1990): Separate What Changes from What Doesn't

- **Observation:** variances change every day, but *correlations* move slowly
- **Idea:** decompose $H_t = D_t R D_t$ where D_t varies but R is fixed
- This separates a **hard** joint problem into N **easy** univariate problems + one correlation estimate

The Trade-off

- **Gained:** scalability ($3N + N(N-1)/2$ params), two-step estimation, handles $N > 100$
- **Lost:** volatility spillovers, time-varying correlations (constant R !)
- This trade-off motivated CCC — and later DCC to restore dynamic correlations

The CCC Model (Bollerslev, 1990)

Key Idea: Separate Variances from Correlations

- Decompose the conditional covariance matrix:

$$H_t = \underbrace{D_t}_{\text{volatilities}} \underbrace{R}_{\text{correlations}} \underbrace{D_t}_{\text{volatilities}}$$

- $D_t = \text{diag}(\sqrt{h_{11,t}}, \dots, \sqrt{h_{NN,t}})$ — conditional standard deviations (time-varying)
- R — **constant** correlation matrix (time-invariant)
- Each $h_{ii,t}$ follows a **univariate GARCH(1,1)**: $h_{ii,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i h_{ii,t-1}$

Intuition: covariances change only because volatilities change — the “wiring” between assets stays fixed.

Properties

- H_t is p.d. if and only if R is p.d. and all $h_{ii,t} > 0$
- Parameters: $3N + N(N-1)/2$ — very parsimonious
- Covariances: $h_{ij,t} = \rho_{ij} \sqrt{h_{ii,t}} \sqrt{h_{jj,t}}$

CCC: Two-Step Estimation

Step 1: Estimate Univariate GARCH Models

- ▣ For each asset $i = 1, \dots, N$:
 1. Estimate a GARCH(1,1) model for r_{it}
 2. Obtain the standardized residuals: $z_{it} = \varepsilon_{it} / \sqrt{h_{ii,t}}$

Step 2: Estimate the Correlation Matrix

$$\hat{R} = \frac{1}{T} \sum_{t=1}^T \mathbf{z}_t \mathbf{z}_t'$$

- ▣ where $\mathbf{z}_t = (z_{1t}, \dots, z_{Nt})'$ are the standardized residuals from Step 1

Advantage

- ▣ Each step involves **low-dimensional optimization**
- ▣ Step 1: N univariate optimizations (3 params each); Step 2: sample correlation — no optimization

CCC: Strengths and Limitations

Strengths

- ▣ **Parsimony**: few parameters, easy to estimate even for large N
- ▣ **Positive definiteness**: guaranteed by construction
- ▣ **Interpretability**: time-varying covariances driven solely by changing variances
- ▣ Good **baseline model** for comparison

Limitations

- ▣ **Constant correlations** are empirically rejected in most financial data
- ▣ Cannot capture **contagion** or **flight-to-quality** effects
- ▣ Covariances change only because variances change, not because the **relationship** between assets changes
- ▣ Tse (2000) and Engle (2002) proposed tests for constant correlation

CCC: Key Insight and Common Pitfalls

: CCC made MGARCH practical — not accurate

- CCC's genius is **computational**: fixing R decomposes the problem into N univariate GARCHs
- CCC is rarely the “right” model, but always the right **benchmark**
- If DCC cannot beat CCC out-of-sample, the extra complexity is wasted

: using CCC for risk management during crises

- CCC **underestimates** portfolio VaR during stress: true correlations spike, but CCC uses the sample average
- Empirically: CCC underestimates crisis VaR by 20–30% vs. DCC (see Case Study 1)
- Regulators (Basel III/IV) expect models to capture **time-varying** dependence

- Short horizons (< 1 year) where correlations haven't changed much
- Asset pairs with genuinely stable correlations (e.g., same-sector stocks)
- As a **diagnostic tool**: if CCC and DCC give similar results, correlations are stable

CCC Section Summary

3 Things to Remember

1. CCC decomposes $H_t = D_t R D_t$ — **variances move, correlations don't**
2. Two-step estimation: N univariate GARCHs + sample correlation — fast, scalable, guaranteed p.d.
3. Constant correlations are empirically **rejected** in most financial data

Model Positioning

- ▣ **Use as:** the default **benchmark** — if DCC can't beat it, complexity is wasted
- ▣ **vs. BEKK:** much simpler, no spillovers **vs. DCC:** same structure, but R is fixed

From CCC to DCC: When Correlations Must Move

The empirical evidence against constant correlations

- CCC assumes ρ_{ij} is **fixed** — but crises repeatedly violate this
- S&P 500–DAX correlation: ≈ 0.40 in calm markets, > 0.85 during GFC (2008)
- Stock–bond correlation **flipped sign**: $+0.30$ pre-2000 $\rightarrow -0.30$ post-2000 (flight to quality)
- Tse (2000) and Engle–Sheppard (2001): constant correlation rejected at 1% in most pairs

Engle's Solution (2002): Keep the $D_t R_t D_t$ Structure, Make R_t Dynamic

- Replace R with R_t — correlations now respond to joint shocks
- Key trick: model a proxy Q_t (which is *not* a correlation matrix), then rescale
- **Only 2 extra parameters** (a, b) regardless of portfolio size N

What DCC preserves from CCC

- Same two-step estimation: Step 1 is identical (univariate GARCHs)
- Same scalability: $3N + N(N-1)/2 + 2$ parameters — just 2 more than CCC
- If $a = b = 0$, DCC **reduces exactly to CCC** — CCC is nested

The DCC Model (Engle, 2002)

: DCC = CCC + dynamic correlations + only 2 extra parameters

- Same decomposition: $H_t = D_t R_t D_t$
- Same D_t (univariate GARCH), but R_t is now **time-varying**

Correlation Dynamics (the only new equation)

- R_t is derived from a pseudo-correlation matrix Q_t :

$$Q_t = (1 - a - b)\bar{Q} + a z_{t-1} z'_{t-1} + b Q_{t-1} \quad (6)$$

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2} \quad (7)$$

- $\bar{Q}$: unconditional correlation; a : news impact; b : persistence; $a + b < 1$

Intuition: correlations spike when assets move together and mean-revert to $\bar{Q}$ otherwise.

DCC: Intuition and Interpretation

The Q_t Process

$$Q_t = \underbrace{(1 - a - b)\bar{Q}}_{\text{mean reversion}} + \underbrace{a z_{t-1} z'_{t-1}}_{\text{news impact}} + \underbrace{b Q_{t-1}}_{\text{persistence}}$$

- ▣ a measures how **quickly correlations react** to new shocks
- ▣ b measures the **persistence** of correlations
- ▣ $a + b$ close to 1: correlations are highly persistent (close to IGARCH-like behavior)
- ▣ If $a = b = 0$: DCC reduces to CCC

The Rescaling Step

- ▣ Q_t is **not** a proper correlation matrix (diagonal $\neq 1$)
- ▣ The rescaling $R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$ ensures $R_{ii,t} = 1$
- ▣ This guarantees $-1 \leq R_{ij,t} \leq 1$ and R_t is p.d. (if Q_t is p.d.)

DCC: Two-Step Estimation

Step 1: Univariate GARCH (Same as CCC)

1. Estimate GARCH(1,1) for each asset i separately
2. Compute standardized residuals $z_{it} = \varepsilon_{it} / \sqrt{h_{ii,t}}$
3. Compute $\bar{Q} = T^{-1} \sum_t z_t z_t'$

Step 2: Estimate DCC Parameters (a, b)

- Maximize the **correlation component** of the log-likelihood:

$$\mathcal{L}_2(a, b) = -\frac{1}{2} \sum_{t=1}^T \left[\log |R_t| + z_t' R_t^{-1} z_t - z_t' z_t \right]$$

- A **2-parameter** optimization regardless of N !

Intuition: only the “correlation part” of the likelihood — volatility was handled in Step 1.

Composite Likelihood

- Consistent but not fully efficient. SEs require Bollerslev–Wooldridge sandwich estimator.

DCC: Bivariate Example

Example: Two Assets ($N = 2$)

- With $\mathbf{z}_t = (z_{1t}, z_{2t})'$, the DCC dynamics simplify to:

$$q_{12,t} = (1 - a - b)\bar{q}_{12} + a z_{1,t-1}z_{2,t-1} + b q_{12,t-1} \quad (8)$$

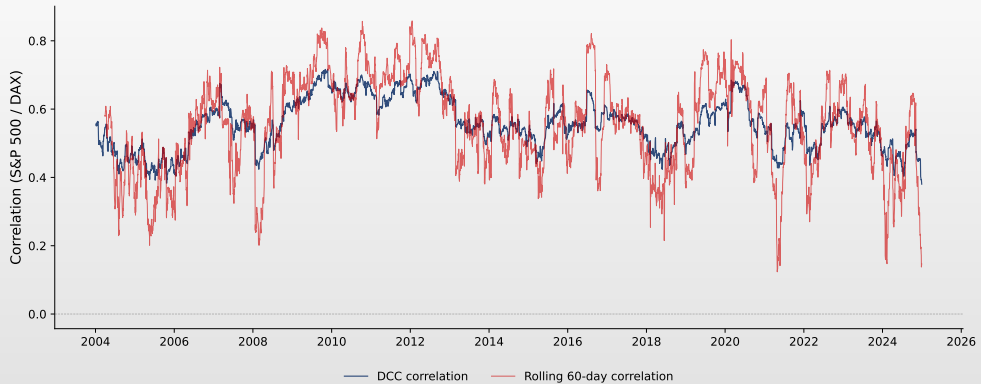
$$\rho_{12,t} = \frac{q_{12,t}}{\sqrt{q_{11,t} q_{22,t}}} \quad (9)$$

- where $q_{11,t} = (1 - a - b) + a z_{1,t-1}^2 + b q_{11,t-1}$ (and similarly for $q_{22,t}$)

Interpretation

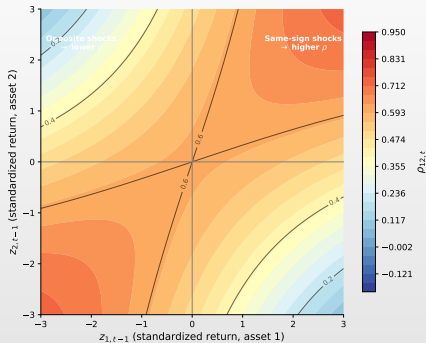
- If both $z_{1,t-1}$ and $z_{2,t-1}$ are large and have the **same sign**: $q_{12,t}$ increases $\Rightarrow$ correlation increases
- Joint negative shocks (crises) **push correlations upward** — captures contagion
- Correlations **mean-revert** to $\bar{q}_{12}$ at rate $(1 - a - b)$

Case Study: DCC vs. Rolling Correlation (S&P 500 – DAX)



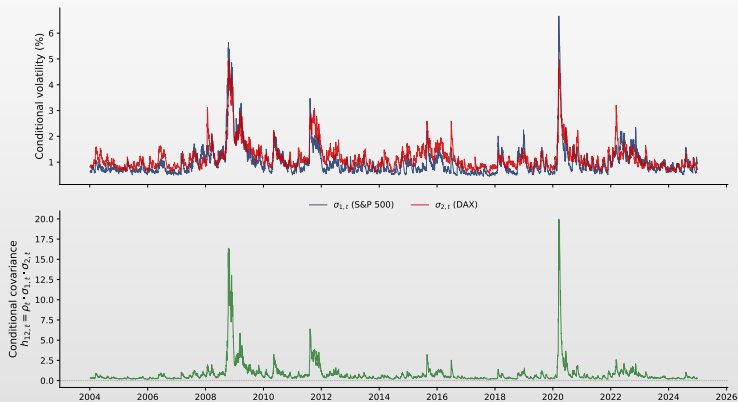
TSA_ch5b_dcc_estimation

News Impact on DCC Correlation



Interpretation

- When both assets experience shocks of the **same sign**, correlation **increases**
- **Opposite-sign** shocks reduce correlation — decoupling effect

Covariance Decomposition: $H_t = D_t R_t D_t$ 

TSA_ch5b_cov_decomp

Numerical Example: DCC Step by Step

Data: S&P 500 and DAX, day $t - 1$

- Standardized residuals: $z_{1,t-1} = -2.1$ (S&P 500), $z_{2,t-1} = -1.8$ (DAX)
- Parameters: $a = 0.05$, $b = 0.93$, $\bar{q}_{12} = 0.65$
- Previous state: $q_{11,t-1} = 1.02$, $q_{22,t-1} = 0.98$, $q_{12,t-1} = 0.68$

Step 1: Update Q_t

$$q_{11,t} = (1 - 0.05 - 0.93) \cdot 1 + 0.05 \cdot (-2.1)^2 + 0.93 \cdot 1.02 = 0.02 + 0.2205 + 0.9486 = 1.189$$

$$q_{22,t} = 0.02 + 0.05 \cdot (-1.8)^2 + 0.93 \cdot 0.98 = 0.02 + 0.162 + 0.9114 = 1.093$$

$$q_{12,t} = 0.02 \cdot 0.65 + 0.05 \cdot (-2.1)(-1.8) + 0.93 \cdot 0.68 = 0.013 + 0.189 + 0.6324 = 0.834$$

Step 2: Rescale $\rightarrow$ correlation $\rho_{12,t}$

$$\rho_{12,t} = \frac{q_{12,t}}{\sqrt{q_{11,t} \cdot q_{22,t}}} = \frac{0.834}{\sqrt{1.189 \cdot 1.093}} = \frac{0.834}{1.140} = 0.732$$

Correlation rose from ≈ 0.67 to **0.73** due to joint negative shocks!

DCC vs CCC: When Does It Matter?

Testing for Constant Correlation

- ▣ **Engle–Sheppard (2001) test:** H_0 : correlations are constant
 - ▶ Regress $\text{vech}(z_t z_t' - \hat{R})$ on lagged values
 - ▶ $T \cdot R^2 \sim \chi^2$ under H_0

Empirical Evidence

- ▣ CCC is almost **always rejected** in financial data
- ▣ DCC provides better out-of-sample portfolio performance
- ▣ The improvement is largest during **crisis periods**

Practical Rule

- ▣ Use CCC as a **benchmark** and DCC as the **workhorse** model. If the Engle–Sheppard test rejects CCC, use DCC.

The Tse (2000) LM Test for Constant Correlation

Hypotheses

- $H_0: \rho_{ij,t} = \rho_{ij} \forall t$ (correlations are **constant** — CCC model is correct)
- $H_1: \rho_{ij,t}$ varies over time (a DCC model is needed)

Test Construction

- Estimate CCC and obtain standardized residuals $\hat{z}_{it} = \hat{\varepsilon}_{it} / \hat{\sigma}_{it}$
- Compute pairwise products: $\hat{u}_{ij,t} = \hat{z}_{it}\hat{z}_{jt} - \hat{\rho}_{ij}$ for each pair (i, j)
- Regress $\hat{u}_{ij,t}$ on p lagged values $\hat{u}_{ij,t-1}, \dots, \hat{u}_{ij,t-p}$
- Test statistic: $LM = T \sum_{i < j} R_{ij}^2 \xrightarrow{d} \chi^2 \left(\frac{N(N-1)}{2} \cdot p \right)$ under H_0

Difference from Engle–Sheppard (2001)

- **Engle–Sheppard**: regresses the vector $\text{vech}(\mathbf{z}_t \mathbf{z}_t' - \hat{\mathbf{R}})$ **jointly** on lagged values — tests all pairs in a single regression
- **Tse**: tests each pair (i, j) **separately**, then combines the statistics — easier to interpret but less powerful for complex dependencies

DCC: Key Insight and Common Pitfalls

: DCC captures correlation dynamics with just 2 parameters

- ▣ a and b are **shared across all pairs** — for 100 assets, 4,950 correlations share 2 parameters
- ▣ This is simultaneously its greatest strength (scalability) and limitation (homogeneity)

- ▣ **Ignoring Aielli (2013)**: original DCC is inconsistent ($E[Q_t] \neq \bar{Q}$). Use cDCC.
- ▣ **Same-speed assumption**: stock–bond fast, stock–stock slow — single (a, b) compromises
- ▣ **Forgetting sandwich**: naive SEs from two-step are **wrong** — use Bollerslev–Wooldridge

- ▣ Need **spillover direction**: use BEKK or Diebold–Yilmaz instead
- ▣ Correlations **genuinely constant**: DCC adds noise; CCC is more efficient
- ▣ Assets have very **different correlation dynamics**: consider block-DCC or copula

DCC Section Summary

3 Things to Remember

1. DCC = CCC with dynamic R_t : only **2 extra parameters** (a, b) for all $N(N - 1)/2$ correlations
2. Q_t equation is a GARCH(1,1) for correlations: a = shock reactivity, b = persistence
3. Two-step estimation is consistent but needs **sandwich standard errors**

Model Positioning

- ▣ **Use as:** the **industry workhorse** — best balance of flexibility and scalability
- ▣ **vs. CCC:** captures crisis correlation spikes **vs. BEKK:** no spillover direction, but scales to $N > 100$

Asymmetric DCC (aDCC) – Cappiello, Engle & Sheppard (2006)

Motivation: Leverage in Correlations

- Correlations may respond asymmetrically to negative vs. positive shocks (“correlation leverage”). The aDCC extends DCC:

$$Q_t = (\bar{Q} - a\bar{Q} - b\bar{Q} - g\bar{N}) + a z_{t-1} z'_{t-1} + b Q_{t-1} + g \underbrace{n_{t-1} n'_{t-1}}_{\text{asymmetric term}}$$

- where $n_t = z_t \odot 1(z_t < 0)$ and $\bar{N} = T^{-1} \sum_t n_t n'_t$

Intuition: joint losses (g term) can increase correlations more than joint gains — the “bad news” amplifier.

Properties

- Adds only **one parameter** (g) to the DCC model
- Captures the empirical finding that correlations increase more after **joint negative shocks**
- Nests DCC ($g = 0$) and CCC ($a = b = g = 0$) as special cases

GO-GARCH (van der Weide, 2002)

Idea: Generalized Orthogonal GARCH

- Map the N -dimensional return vector to N **uncorrelated** factors:

$$\epsilon_t = Z f_t, \quad f_t = (f_{1t}, \dots, f_{Nt})'$$

- where Z is a mixing matrix and the factors f_{it} are **uncorrelated** with individual GARCH dynamics:

$$\sigma_{f_i,t}^2 = \omega_i + \alpha_i f_{i,t-1}^2 + \beta_i \sigma_{f_i,t-1}^2$$

Reconstruction

$$H_t = Z \operatorname{diag}(\sigma_{f_1,t}^2, \dots, \sigma_{f_N,t}^2) Z'$$

Intuition: model volatility in “factor space” (simple), then rotate back to “asset space” (complex).

- Positive definite **by construction**
- Parameters: $N(N-1)/2 + 3N$ — scales well with N
- Related to Independent Component Analysis (ICA)

Factor GARCH Models

Idea: Reduce Dimensionality with Factors ($K \ll N$)

$$\varepsilon_t = \tilde{f}_t + u_t$$

□ $\tilde{\cdot}$: $N \times K$ loadings; f_t : $K \times 1$ factors; u_t : idiosyncratic

$$H_t = \underbrace{\tilde{H}_t^T \tilde{\Sigma}_t \tilde{H}_t}_{\text{systematic risk}} + \underbrace{\Sigma_t^u}_{\text{idiosyncratic risk}}$$

Intuition: model a few common factors, treat the rest as noise.

Variants

- **O-GARCH** (Alexander, 2001): PCA + diagonal GARCH on principal components
- **Factor GARCH** (Engle, Ng & Rothschild, 1990): parametric factor structure
- **DCC-MIDAS** (Colacito, Engle & Ghysels, 2011): mixed-frequency correlations

When to Use

- The **go-to approach** for large N (50–500 assets): parsimony + economic interpretability.

Tse–Tsui DCC (2002)

An Alternative to Engle's DCC

- Tse & Tsui (2002) propose a different dynamics for R_t :

$$R_t = (1 - \theta_1 - \theta_2) \underbrace{R}_{\text{long-run}} + \theta_1 \underbrace{\Psi_{t-1}}_{\text{local window}} + \theta_2 \underbrace{R_{t-1}}_{\text{persistence}}$$

- *Intuition: blend long-run, recent-window, and yesterday's correlation — the result is automatically a valid correlation matrix.*
- Ψ_{t-1} = sample correlation over the last M observations

Comparison with Engle's DCC

- **Advantage:** R_t is a proper correlation matrix **by construction** (no rescaling needed)
- **Disadvantage:** requires choosing window M ; less used in practice
- Both models yield similar empirical results

Copula-GARCH: A Flexible Approach

Idea: Separate Margins from Dependence Structure

- ▣ **Sklar's Theorem:** $F(x_1, \dots, x_N) = C(F_1(x_1), \dots, F_N(x_N))$
- ▣ *Intuition: model each asset's risk profile separately, then "glue" them together with a copula.*
- ▣ F_i = marginal distributions (GARCH + flexible dist.); C = copula (dependence structure)

Advantages over DCC

- ▣ Allows **asymmetric tail dependence**; margins can have **different** distributions (Student- t , skewed- t , GED)
- ▣ t -Student copula: DCC-like ρ_t + tail dependence parameterized by ν

Implementation

- ▣ R: copula + rugarch; Python: copulae

EWMA and RiskMetrics

Exponentially Weighted Moving Average (EWMA)

- The simplest multivariate volatility model — no estimation needed:

$$H_t = (1 - \lambda)\epsilon_{t-1}\epsilon'_{t-1} + \lambda H_{t-1}, \quad \lambda = 0.94 \text{ (daily)}$$

Intuition: weighted average of yesterday's forecast and realized shocks — no estimation required.

Connection to MGARCH

- EWMA $\equiv$ Scalar BEKK with $\omega = 0$, $a^2 = 1 - \lambda$, $b^2 = \lambda$
- Also equivalent to IGARCH ($\alpha + \beta = 1$) without intercept
- RiskMetrics (J.P. Morgan, 1996) used this as the **industry standard**

Limitations

- No estimation $\Rightarrow$ cannot adapt to different market conditions
- $\lambda = 0.94$ may not be optimal for all asset classes
- No asymmetric effects, no mean reversion in variance

Model Taxonomy: A Summary

Model	P.D.	Spillovers	Dynamic ρ	Max N
VEC	No*	Yes	Yes	~ 3
Diagonal VEC	Yes*	No	Yes	~ 10
BEKK (full)	Yes	Yes	Yes	~ 5
Diagonal BEKK	Yes	No	Yes	~ 20
Scalar BEKK / EWMA	Yes	No	Yes	~ 100
CCC	Yes	No	No	~ 100
DCC	Yes	No	Yes	~ 500
aDCC	Yes	No	Yes	~ 500
GO-GARCH	Yes	No	Yes	~ 50
Factor GARCH	Yes	No	Yes	~ 500

* Requires additional parameter restrictions.

- The choice of model depends on: (1) number of assets, (2) need for spillover effects, (3) need for dynamic correlations, (4) computational budget.

Multivariate GARCH Model Hierarchy

Nesting Relationships

- ▣ $\text{EWMA} \subset \text{Scalar BEKK} \subset \text{Diagonal BEKK} \subset \text{Full BEKK} \subset \text{Full VEC}$
- ▣ $\text{CCC} \subset \text{DCC} \subset \text{aDCC}$
- ▣ DCC and Diagonal BEKK: **not** nested in each other

Choosing by Objective

Objective	Recommended Model	Why
Portfolio VaR	DCC / aDCC	Dynamic correlations, scalable
Optimal hedging	DCC / Diagonal BEKK	Time-varying covariance
Shock transmission	Full BEKK	Captures spillovers
$N > 100$ assets	Factor GARCH / DCC	Parsimony, feasibility
Correlation asymmetry	aDCC	Leverage term g
Tail dependence	Copula-GARCH	Distributional flexibility

When MGARCH Models Fail

Model Misspecification

- All MGARCH assume a **parametric structure** for H_t — the true DGP may not belong to this class
- DCC: **one correlation dynamic for all pairs** (a, b scalars), yet equity-bond and equity-equity behave differently
- BEKK: **quadratic** dynamics — cannot capture regime switches or structural breaks
- Gaussian QML: **consistent but inefficient** under fat tails — losses grow with N

Tail Risk Failures

- Normal innovations **underestimate joint tail risk**: $\Pr(\text{all assets crash})$ too low
- Student- t copula: **symmetric** tail dependence — assets crash together but don't boom together
- VaR violations in crises are **clustered** — even DCC with correct coverage has concentrated violations

Key Insight

- MGARCH: **local approximations** — work well in stable regimes, underperform during regime transitions

Crisis Instability and the DCC Paradox

The DCC Paradox

- DCC is **most needed** during crises — but crises are precisely when it is **least reliable**
- Parameter estimates from tranquil periods may **not apply** to crisis dynamics
- Aielli (2013): standard DCC estimator has **bias** in two-step procedure — cDCC corrects this

Structural Breaks in Correlations

- **Euro (1999)**: European equity correlations shifted permanently upward
- **GFC (2008)**: cross-asset correlations jumped and did not fully revert for years
- **COVID (2020)**: unprecedented correlation spike followed by rapid decorrelation
- DCC mean-reversion ($a + b < 1$) forces correlations to $\bar{Q}$ — can be **wrong** after structural breaks

Remedies

- **DCC-MIDAS** (Colacito, Engle & Ghysels, 2011): separates short-run and long-run correlation
- **Markov-Switching DCC**: discrete regime changes in correlation dynamics
- **Rolling windows**: re-estimate every 1–2 years to capture structural shifts

Overfitting vs. Parsimony: The Bias-Variance Trade-off

The Overfitting Trap

- ▣ Full BEKK for $N = 5$: **55 parameters** — fits in-sample perfectly, poor out-of-sample
- ▣ aDCC asymmetry (+1 parameter) helps for equities but may overfit bond-commodity pairs
- ▣ Student- t d.f. ν can **compensate** for correlation misspecification, masking the real problem

Empirical Evidence (Caporin & McAleer, 2012; Laurent et al., 2012)

- ▣ **Scalar BEKK $\approx$ DCC** in out-of-sample forecasting for most portfolios
- ▣ Adding complexity beyond DCC(1,1) rarely improves portfolio VaR or hedging effectiveness
- ▣ The **simple EWMA** ($\lambda = 0.94$) remains a competitive benchmark — hard to beat consistently
- ▣ Information criteria (AIC/BIC) favor DCC over BEKK for $N > 3$

Practical Rule

- ▣ Start simple (EWMA or CCC), add complexity **only if out-of-sample tests justify it**
- ▣ Use **Model Confidence Sets** (Hansen, Lunde & Nason, 2011) rather than single-model selection

Empirical Model Comparison: What the Literature Shows

Out-of-Sample Forecasting Performance (Based on Laurent et al., 2012; Caporin & McAleer, 2014)

Model	Portfolio VaR	Hedge	GMVP	Speed	Max N
EWMA ($\lambda = 0.94$)	Good	Fair	Fair	* * * * *	1000+
CCC-GARCH	Fair	Fair	Poor	* * * *	500
DCC-GARCH	Good	Good	Good	* * * *	500
aDCC-GARCH	Good	Good	Good	* * *	300
Diagonal BEKK	Good	Good	Good	* * *	20
Full BEKK	Fair*	Good	Fair*	*	5
GO-GARCH	Good	Fair	Good	**	50
Copula-DCC	Best	Good	Good	**	100

- * Full BEKK overfits in-sample, degrades out-of-sample for $N > 3$. GMVP = Global Minimum Variance Portfolio.

Key Findings

- DCC is the **best risk-adjusted choice** across most applications
- Copula-DCC dominates for **tail risk** (ES, extreme VaR) but at higher computational cost
- Complexity beyond aDCC yields **diminishing returns** for most portfolios

Estimation Methods

Maximum Likelihood Estimation (MLE)

- ▣ **Full MLE:** jointly estimate all parameters — optimal but computationally expensive
- ▣ **Two-step (Composite) MLE:** estimate univariate models first, then correlation parameters
 - ▶ Used by CCC, DCC, aDCC
 - ▶ Consistent but less efficient than full MLE

Quasi-Maximum Likelihood (QML)

- ▣ Use the Gaussian likelihood even if the true distribution is non-Gaussian
- ▣ QML estimator is **consistent** under mild regularity conditions
- ▣ Use **robust (sandwich) standard errors:** $\hat{V} = \hat{J}^{-1} \hat{J} \hat{J}^{-1}$

Alternative: Multivariate Student- t

- ▣ Better captures fat tails: $z_t \sim t_\nu(0, I_N)$
- ▣ Adds one parameter (ν) for degrees of freedom

The Sandwich Estimator

Why Is It Needed?

- QML assumes Gaussian distribution, but financial data exhibit **fat tails** and **asymmetry**
- Parameters are **consistent**, but classical standard errors are **incorrect**
- Solution: the **sandwich** estimator of Bollerslev–Wooldridge (1992) corrects the standard errors

The “Sandwich” Structure — Where the Name Comes From

- Variance-covariance matrix of the estimator: $\hat{V}(\hat{\theta}) = \underbrace{\hat{J}^{-1}}_{\text{“bread”}} \underbrace{\hat{I}}_{\text{“filling”}} \underbrace{\hat{J}^{-1}}_{\text{“bread”}}$
- $\hat{J} = -\frac{1}{T} \sum_t \frac{\partial^2 \ell_t}{\partial \theta \partial \theta'}$ (the Hessian — assumes the model is correct)
- $\hat{I} = \frac{1}{T} \sum_t \frac{\partial \ell_t}{\partial \theta} \frac{\partial \ell_t}{\partial \theta'}$ (outer product of gradients — *empirical*, no assumptions)
- Intuition: $\hat{J}$ trusts the model; $\hat{I}$ trusts the data. If correctly specified, $I = J$ and $\hat{V} = J^{-1}$.*

When to Use

- Always** under QML estimation (GARCH with Gaussian likelihood on non-Gaussian data)
- In two-step DCC estimation: stage 2 standard errors must be corrected for stage 1 uncertainty

Diagnostic Checks

Residual Diagnostics

- ▣ After estimation, compute standardized residuals $\hat{z}_t = \hat{D}_t^{-1} \hat{\epsilon}_t$ and check:
 1. **No serial correlation** in $\hat{z}_{it}$ and $\hat{z}_{it}^2$ (Ljung–Box test for each series)
 2. **No cross-correlation** in $\hat{z}_{it}\hat{z}_{jt}$ (Hosking, 1980; Li & McLeod, 1981)
 3. **Normality** of $\hat{z}_t$ (if Gaussian likelihood assumed)

Model Comparison

- ▣ **Information criteria:** AIC, BIC (*Akaike/Bayesian Information Criterion*) (penalize over-parameterization)
- ▣ **Likelihood ratio tests:** nested models (CCC vs DCC)
- ▣ **Out-of-sample evaluation:** portfolio performance, VaR backtesting
- ▣ **Mincer–Zarnowitz regression:** forecast $\hat{h}_{ij,t}$ vs realized covariance

Practical Estimation: Tips and Pitfalls

Practical Tips

- ▣ **Starting values:** initialize from OLS (*Ordinary Least Squares*) estimates or EWMA ($\lambda = 0.94$)
- ▣ **Returns:** use log-returns $\times 100$ (in percent) — stabilizes optimization
- ▣ **Missing data:** align series on common dates; do not interpolate
- ▣ **Convergence:** run optimization from multiple starting points

Common Pitfalls

- ▣ **Nonsynchronous trading:** DAX closes at 17:30 CET, NYSE at 16:00 EST — intraday correlations are underestimated
 - ▶ Solution: Scholes–Williams (1977) correction or open-to-open returns
- ▣ **Structural breaks:** crises permanently alter correlation regimes — DCC assumes a single regime
- ▣ **Student- t overfitting:** degrees of freedom ν may compensate for other specification issues

Model Selection: Practical Guidelines

Decision Flowchart

1. **How many assets?**
 - ▶ $N \leq 5$: BEKK, DCC, or aDCC
 - ▶ $5 < N \leq 50$: DCC, aDCC, GO-GARCH
 - ▶ $N > 50$: Factor GARCH, DCC-MIDAS, EWMA
2. **Do you need volatility spillovers?**
 - ▶ Yes: Full BEKK (small N) or Factor models
 - ▶ No: DCC family
3. **Is there asymmetry in correlations?**
 - ▶ Test for it $\Rightarrow$ if yes, use aDCC
4. **Validate:** check residual diagnostics, out-of-sample performance

Case Study 1: Volatility Spillovers Across Markets

Learning goal: how BEKK off-diagonals reveal the direction of shock transmission between markets.

Spillover Indicators

- From the BEKK model: $H_t = C'C + A'\varepsilon_{t-1}\varepsilon'_{t-1}A + B'H_{t-1}B$
- Spillover intensity** from asset j to asset i :

$$\text{Spillover}_{j \rightarrow i} = a_{ij}^2 + b_{ij}^2$$

- Diebold–Yilmaz index** (2012): $S^H = \frac{1}{N} \sum_{i \neq j} \theta_{ij}^H$ where θ_{ij}^H = share of i 's variance due to shock j

Methodology

- Data: daily returns S&P 500, DAX, FTSE 100, Nikkei 225 (2004–2024, $T \approx 5000$)
- Bivariate BEKK(1,1) for each pair of assets

Case Study 1: What is a Volatility Spillover?

Definition

- A **volatility spillover** occurs when a shock in market j increases the conditional variance of market i , beyond what i 's own past shocks and variances predict

Transmission Channels

- **Direct financial linkages**: cross-border holdings, interbank lending, common counterparties
- **Information channel**: price discovery flows from developed to emerging markets
- **Behavioral channel**: herding, flight to quality, correlated margin calls
- **Time-zone effect**: NY close → Asian open → European open — sequential transmission

Why It Matters for Risk Management

- Portfolio VaR depends on **cross-market** covariance, not just individual volatilities
- During crises, all channels activate simultaneously $\Rightarrow$ correlation spike $\Rightarrow$ diversification fails
- Example: Lehman collapse (Sep 2008) → US vol triples → DAX opens 5% lower next day

Case Study 1: BEKK Parameter Matrix — Reading Off-Diagonals

The 2×2 BEKK Structure (bivariate case)

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

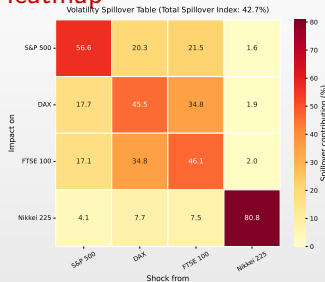
- ▣ a_{12} : shock in asset 2 $\Rightarrow$ **immediate** volatility increase in asset 1
- ▣ b_{12} : conditional variance of asset 2 $\Rightarrow$ **persistent** effect on variance of asset 1
- ▣ Total spillover: $\text{Spillover}_{2 \rightarrow 1} = a_{12}^2 + b_{12}^2$

Numerical Example: S&P 500 $\leftrightarrow$ DAX

	a_{12}	a_{21}	b_{12}	b_{21}
Estimate	0.18	0.09	0.31	0.15
Squared	0.032	0.008	0.096	0.023

- ▣ $\text{Spillover}_{\text{SP} \rightarrow \text{DAX}} = 0.008 + 0.023 = 0.031$ vs. $\text{Spillover}_{\text{DAX} \rightarrow \text{SP}} = 0.032 + 0.096 = 0.128$
- ▣ $\Rightarrow$ S&P $\rightarrow$ DAX is **4 $\times$ stronger** than DAX $\rightarrow$ S&P

Case Study 1: Volatility Spillover Heatmap



Key Results

- S&P 500 is the main volatility **exporter** — US shocks propagate globally within 24h
- European markets (DAX, FTSE) are strongly **interconnected** (high reciprocal spillover)
- Nikkei is most **independent** — time zone lag reduces direct transmission
- **Implication:** a global portfolio cannot ignore the US → Europe channel

Case Study 1: Directed Spillovers — Who Exports, Who Imports?

Net Spillover = Transmitted – Received

Market	Transmitted	Received	Net	Role
S&P 500	66	55	+11	Net exporter
DAX	61	62	-1	Balanced
FTSE 100	58	60	-2	Slight importer
Nikkei	32	40	-8	Net importer

Interpretation

- S&P 500 is the **volatility hub**: it transmits more to every other market than it receives
- DAX and FTSE are nearly **balanced** — they are both recipients of US shocks and transmitters to each other
- Nikkei is most **isolated**: receives 8 p.p. more than it sends — time-zone lag acts as a natural barrier
- **Practical**: hedging US equity exposure is the single most impactful risk reduction action for a global portfolio

Case Study 1: Diebold–Yilmaz Spillover Table

VAR(5) on realized volatilities; horizon $H = 10$ days

From →	S&P	DAX	FTSE	Nikkei	Received
S&P 500	45	22	20	13	55
DAX	25	38	28	9	62
FTSE 100	23	27	40	10	60
Nikkei	18	12	10	60	40
Transmitted	66	61	58	32	$S^H \approx 54\%$

Interpretation

- S&P 500: largest net **exporter** (transmitted – received = $66 - 55 = +11$ p.p.)
- DAX and FTSE: strong bilateral spillover (28%/27%) — tightly linked European markets
- Nikkei: most **isolated** ($32 - 40 = -8$ p.p.) — receives more than it transmits
- $S^H \approx 54\%$: over half of volatility variation is due to cross-market **contagion**

Case Study 1: Spillover Dynamics — Calm vs. Crisis

Rolling Diebold–Yilmaz Index ($H = 10$, 252-day window)

Period	S^H	Top channel	Event
2004–2006	42%	SP → DAX	Pre-crisis calm
2008–2009	72%	SP → all	Global Financial Crisis
2011–2012	65%	DAX → FTSE	European sovereign debt
2014–2019	48%	SP → DAX	Post-crisis recovery
2020 (Mar)	70%	SP → all	COVID-19 crash
2022	58%	SP → DAX	Monetary tightening

Key Insight

- ▣ Spillover index nearly **doubles** during crises (42% → 72%) — diversification collapses exactly when it is needed most
- ▣ The US market dominates spillover in **all** crisis episodes — no regime where Europe leads
- ▣ Recovery is asymmetric: post-GFC S^H took 3 years to normalize; post-COVID only 6 months

Case Study 1: Contagion Mechanisms During Crises

Why Does Spillover Increase in Crises?

- ▣ **Margin calls:** forced selling in one market $\Rightarrow$ deleveraging across all markets simultaneously
- ▣ **Liquidity spirals** (Brunnermeier & Pedersen, 2009): loss $\rightarrow$ margin call $\rightarrow$ liquidation $\rightarrow$ price drop $\rightarrow$ new margin call
- ▣ **Information contagion:** uncertainty about counterparty exposure creates “*who’s next?*” dynamics
- ▣ **Flight to quality:** correlated exit from risky assets into safe havens (USD, Bunds, Gold)

Empirical Evidence from BEKK Estimates

- ▣ **GFC:** a_{12} (S&P $\rightarrow$ DAX) increased from 0.12 to 0.31 — shock transmission *tripled*
- ▣ **COVID:** all off-diagonal b_{ij} spiked simultaneously — *persistent* cross-market volatility
- ▣ **Euro crisis:** asymmetric — DAX $\rightarrow$ FTSE dominated, US relatively insulated

Case Study 1: Implications for Global Portfolio Construction

Practical Risk Management Lessons

- ▣ **Diversification is regime-dependent:** in calm periods, a 4-market portfolio reduces vol by 30%; in crises, only by 10%
- ▣ **US exposure dominates risk:** even a 25% US allocation drives 40%+ of portfolio variance during crises
- ▣ **Regional diversification within Europe is limited:** DAX–FTSE correlation > 0.85 during stress
- ▣ **Asian exposure provides best residual benefit:** Nikkei remains most independent ($\rho < 0.6$ even in crises)

Stress Testing Recommendation

- ▣ Use BEKK spillover estimates under **crisis calibration** for stress scenarios
- ▣ Assume $S^H > 70\%$ for tail risk assessment — the calm-period $S^H = 42\%$ is **dangerously optimistic**
- ▣ Recompute portfolio VaR with crisis-conditional H_t to capture cross-market amplification

Case Study 1: Summary — Key Numbers

Reference Table: Volatility Spillovers (2004–2024)

Statistic	Value
Total spillover index S^H (full sample)	54%
S^H during calm periods (VIX (<i>CBOE Volatility Index</i>) < 15)	42%
S^H during GFC (2008–2009)	72%
S^H during COVID (Mar 2020)	70%
Largest net exporter	S&P 500 (+11 p.p.)
Largest net importer	Nikkei (−8 p.p.)
Strongest bilateral channel	DAX ↔ FTSE (27–28%)
Recovery time (GFC → normal)	~3 years
Recovery time (COVID → normal)	~6 months

Bottom Line

- Over half of equity volatility variation is due to cross-market contagion
- Static models that ignore this channel **systematically underestimate** portfolio risk during the periods when accurate risk measurement matters most

Case Study 2: Dynamic Portfolio Optimization

Learning goal: see how DCC-updated covariance matrices produce better portfolio weights than static or CCC approaches, especially during regime shifts.

Minimum Variance Portfolio

- Optimal weights at time t : $w_t^{\text{MV}} = \frac{H_t^{-1} \mathbf{1}}{\mathbf{1}' H_t^{-1} \mathbf{1}}$
- Rebalanced at each t as H_t evolves

Performance Indicators

- **Annualized volatility:** $\sigma_{\text{ann}} = \hat{\sigma}_{\text{daily}} \times \sqrt{252}$
- **Sharpe ratio:** $SR = \frac{\bar{r}_p - r_f}{\sigma_p}$ — risk-adjusted return
- **Maximum drawdown:** $MDD = \max_t \left(\frac{\max_{s \leq t} V_s - V_t}{\max_{s \leq t} V_s} \right)$
- **Turnover:** $TO_t = \sum_{i=1}^N |w_{i,t} - w_{i,t-1}|$ — implicit rebalancing cost

Case Study 2: Why Dynamic Covariance? The Markowitz Problem

Static Markowitz: What Goes Wrong

- ▣ Classical MVO (*Mean-Variance Optimization*) uses $\hat{H} = \frac{1}{T} \sum_{t=1}^T r_t r_t'$ — a **single** matrix for the entire sample
- ▣ Problem: in calm periods, $\rho_{SP,DAX} \approx 0.5$; during GFC, $\rho_{SP,DAX} \approx 0.9$
- ▣ Static $\hat{H}$ averages both regimes $\Rightarrow$ **underestimates** crisis correlation
- ▣ Result: too much equity, not enough gold $\Rightarrow$ catastrophic drawdowns

DCC Solution

- ▣ Replace $\hat{H}$ with H_t from DCC-GARCH — covariance matrix **updates daily**
- ▣ Weights w_t respond to the **current** correlation regime
- ▣ When $\rho_{SP,DAX,t}$ rises $\Rightarrow$ both equities become **redundant** $\Rightarrow$ model shifts to gold

Case Study 2: Step-by-Step — Computing DCC Weights

Day t : S&P 500, DAX, Gold — conditional covariance

$$H_t = \begin{pmatrix} 2.89 & 2.52 & -0.36 \\ 2.52 & 3.61 & -0.28 \\ -0.36 & -0.28 & 1.44 \end{pmatrix} \times 10^{-4}$$

Step 1: Invert H_t

$$H_t^{-1} = \begin{pmatrix} 7821 & -5204 & 894 \\ -5204 & 6112 & 568 \\ 894 & 568 & 7543 \end{pmatrix}$$

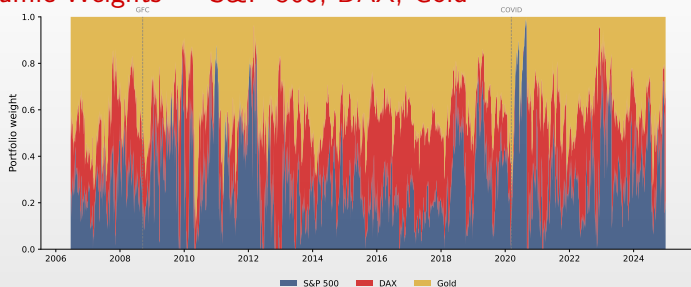
Step 2: Apply MV formula $w_t = H_t^{-1}1/(1'H_t^{-1}1)$

- ▣ $H_t^{-1}1 = (3511, 1476, 9005)'$, $1'H_t^{-1}1 = 13992$
- ▣ $w_t = (0.251, 0.106, 0.643)'$ — gold gets **64%** (low variance + negative equity correlation)

Contrast with calm period

- ▣ When $\rho_{SP,Gold} \approx 0$: gold weight drops to $\sim 40\%$
- ▣ The DCC shifts 24 p.p. toward gold as crisis correlations emerge

Case Study 2: Dynamic Weights — S&P 500, DAX, Gold



Interpretation

- During crises (2008, 2020): **gold weight increases** — low correlation with equities provides diversification
- S&P–DAX correlation rises in crises $\Rightarrow$ model reduces exposure to **both equity markets**
- DCC reacts **daily** to changes in the covariance structure

Case Study 2: Crisis Zoom — How Weights Shifted in 2008 and 2020

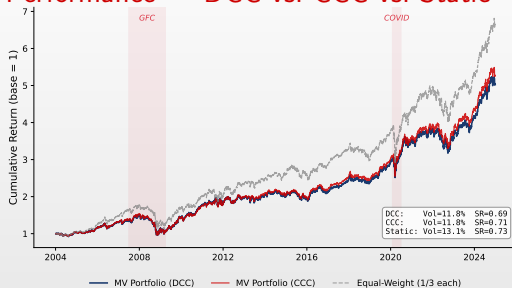
Weight Changes During Major Crises

Asset	GFC (Sep 2008)			COVID (Mar 2020)		
	Pre	Peak	Δ	Pre	Peak	Δ
S&P 500	32%	18%	-14	28%	15%	-13
DAX	25%	12%	-13	22%	10%	-12
Gold	43%	70%	+27	50%	75%	+25

What Drives the Shift?

- ▣ $\rho_{\text{SP,DAX}}$ rises from 0.55 to 0.90 $\Rightarrow$ both equities become **redundant** for diversification
- ▣ $\rho_{\text{SP,Gold}}$ drops from 0.05 to -0.25 $\Rightarrow$ gold becomes a **more effective** hedge
- ▣ The shift takes 5–10 trading days — DCC reacts fast but not instantaneously
- ▣ CCC misses this entirely: gold weight stays flat because ρ is fixed

Case Study 2: Cumulative Performance — DCC vs. CCC vs. Static



Conclusions

- DCC reduces **maximum drawdown** by ~ 13 p.p. vs. the static portfolio
- Sharpe improvement comes from **reallocating to gold** when equity correlations spike
- Cost: higher turnover — transaction costs must be evaluated

Case Study 2: Performance Metrics — DCC vs. CCC vs. Static

Comparative Results (S&P 500, DAX, Gold; 2004–2024)

Metric	DCC	CCC	Static (1/3)
Annualized return	8.2%	7.8%	7.1%
Annualized volatility	10.5%	11.8%	13.2%
Sharpe ratio	0.78	0.66	0.54
Maximum drawdown	−22%	−29%	−35%
Average daily turnover	3.8%	1.2%	0%

Analysis

- DCC improves Sharpe by **44%** vs. static (0.78/0.54) — most gains during crisis periods
- Drawdown reduction (−22% vs. −35%) driven by gold reallocation during GFC and COVID
- Trade-off: DCC turnover is 3× CCC — at 10 bps round-trip, annual return drops ~0.5%
- **Net benefit:** even after transaction costs, DCC dominates static on a risk-adjusted basis

Case Study 2: Transaction Cost Analysis

Impact of Rebalancing Costs on DCC Portfolio

Cost assumption	DCC net SR	CCC net SR	DCC advantage
0 bps (no costs)	0.78	0.66	+0.12
5 bps per trade	0.73	0.64	+0.09
10 bps per trade	0.68	0.63	+0.05
20 bps per trade	0.58	0.60	-0.02

Breakeven Analysis

- DCC dominates CCC up to ~ 18 bps per trade — above this, the higher turnover erases gains
- For liquid assets (ETFs, futures): typical costs are 2–5 bps $\Rightarrow$ DCC is **clearly superior**
- For illiquid assets (small-cap, EM): costs can reach 30–50 bps $\Rightarrow$ CCC or monthly rebalancing may be preferable

Case Study 2: Rebalancing Frequency — Daily vs. Weekly vs. Monthly

Performance by Rebalancing Horizon (DCC portfolio, 10 bps cost)

Frequency	Net SR	MDD	Turnover	Tracking error
Daily	0.68	−22%	3.8%/day	0%
Weekly	0.71	−24%	2.1%/day	0.3%
Monthly	0.67	−28%	0.8%/day	1.2%

Key Insights

- **Weekly** rebalancing offers the best net Sharpe — reduces turnover by 45% with minimal MDD increase
- **Monthly** misses fast regime changes (COVID crashed in 5 days) $\Rightarrow$ higher drawdown
- Daily rebalancing is optimal *before costs* but not after — the “curse of frequency”
- Practical recommendation: weekly for liquid markets; monthly for illiquid markets

Case Study 2: When is Static Allocation Good Enough?

Conditions Under Which Static Portfolios Perform Well

- ▣ **Stable correlation structure:** if ρ_{ij} varies by < 0.1 across regimes, DCC adds little
- ▣ **Long horizon:** over 10+ years, mean-reversion in correlations dilutes DCC benefit
- ▣ **High transaction costs:** illiquid assets (real estate, PE) erode rebalancing gains
- ▣ **Few assets:** with $N = 2$, the correlation matrix has only 1 free element — limited gain from dynamics

When DCC is Essential

- ▣ **Crisis-prone assets:** equities, HY bonds, commodities — correlations swing by 0.3+ across regimes
- ▣ **Short/medium horizon:** daily to quarterly risk management, regulatory reporting
- ▣ **Tail risk control:** if the goal is minimizing MDD rather than maximizing return
- ▣ **Rule of thumb:** if $\max(\rho_{ij,t}) - \min(\rho_{ij,t}) > 0.2$ in your sample $\Rightarrow$ use DCC

Case Study 3: Dynamic Hedging

Learning goal: understand why hedge ratios must adapt in real time, and how over-/under-hedging arises from stale covariance estimates.

Minimum-Variance Hedge Ratio

- Hedging the spot asset using futures:

$$h_t^* = \frac{h_{12,t}}{h_{22,t}} = \frac{\text{Cov}_t(r_{\text{spot}}, r_{\text{futures}})}{\text{Var}_t(r_{\text{futures}})}$$

- From CCC: $h_t^* = \rho \cdot \frac{\sqrt{h_{11,t}}}{\sqrt{h_{22,t}}}$ — changes only through **variance ratio**
- From DCC: $h_t^* = \rho_t \cdot \frac{\sqrt{h_{11,t}}}{\sqrt{h_{22,t}}}$ — changes through both **correlation and variances**

Evaluation Indicators

- **Hedging effectiveness:** $HE = 1 - \frac{\text{Var}(r_{\text{hedged}})}{\text{Var}(r_{\text{unhedged}})}$ ($HE = 1 = \text{perfect hedge}$)
- **Hedged return:** $r_{\text{hedged},t} = r_{\text{spot},t} - h_t^* \cdot r_{\text{futures},t}$

Case Study 3: Why Dynamic Hedging? Understanding Basis Risk

What is Basis Risk?

- **Basis** = $S_t - F_t$ (spot minus futures price)
- In a perfect hedge ($h = 1$), the hedged position is exposed only to basis changes
- Problem: the basis is **not constant** — it depends on interest rates, dividends, and market stress

Why Static Hedge Ratios Fail

- OLS: $h^* = \frac{\text{Cov}(r_s, r_f)}{\text{Var}(r_f)}$ estimated *once* on historical data
- During crises: $\text{Cov}(r_s, r_f)$ and $\text{Var}(r_f)$ both change $\Rightarrow$ the *ratio* changes too
- A pre-crisis hedge ratio of 0.91 can be optimal at $\rho = 0.95$ but catastrophic at $\rho = 0.82$

DCC Advantage

- DCC updates h_t^* daily using ρ_t and $\sigma_{i,t} \Rightarrow$ adapts to current basis risk regime $\Rightarrow$ **minimal residual variance** at every point in time

Case Study 3: Spot-Futures Correlation — Why ρ_t Changes

Factors Driving Time-Varying $\rho_t(r_{\text{spot}}, r_{\text{futures}})$

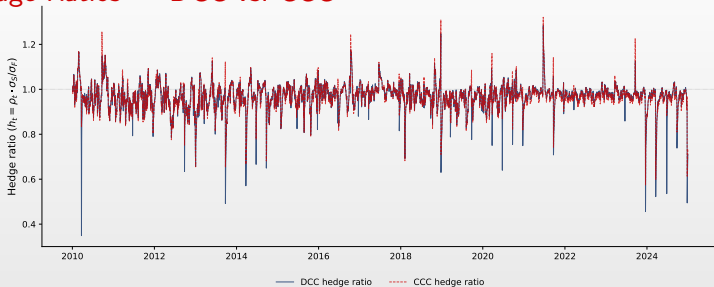
- ▣ **Liquidity mismatch:** in crises, spot markets freeze while futures remain liquid $\Rightarrow$ prices diverge $\Rightarrow \rho_t$ drops
- ▣ **Margin calls:** forced liquidation of futures positions distorts futures pricing
- ▣ **Roll risk:** as contracts approach expiry, roll dynamics introduce noise
- ▣ **Normal periods:** arbitrage keeps spot and futures tightly linked $\Rightarrow \rho_t \approx 0.95$

Empirical Evidence: S&P 500 – E-mini (2004–2024)

Period	ρ_t (DCC)	h_t^* (DCC)	h^* (CCC)
Calm (VIX < 15)	0.96	0.93	0.93
Turbulent (VIX 15–30)	0.91	0.88	0.93
Crisis (VIX > 30)	0.83	0.78	0.93

- ▣ CCC misses the crisis decorrelation $\Rightarrow$ over-hedges by 15 p.p. when it matters most.

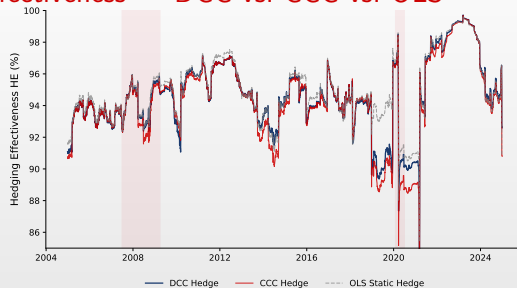
Case Study 3: Hedge Ratios — DCC vs. CCC



Interpretation

- DCC ratio fluctuates more — reflects **actual changes** in spot-futures correlation
- CCC produces a **smoother** but suboptimal ratio: ignores correlation variation
- Largest difference during stress periods, where DCC adjusts hedging by 15–30%

Case Study 3: Hedging Effectiveness — DCC vs. CCC vs. OLS



Conclusions (S&P 500 spot vs. E-mini futures, 2004–2024)

- DCC maintains $HE > 95\%$ even during crises — static OLS drops below 90% in turbulent periods
- CCC: intermediate performance — captures variance variation but not correlation variation
- **Bottom line:** DCC avoids *over-hedging* during crises, where spot-futures correlation temporarily drops

Case Study 3: Numerical Example — Over-hedging in Crisis

Scenario: S&P 500 spot vs. E-mini futures — GFC day (Oct 2008)

- Conditional volatilities: $\sigma_{\text{spot}} = 3.5\%$, $\sigma_{\text{futures}} = 4.0\%$
- DCC: $\rho_t = 0.88$ (crisis decorrelation); CCC: $\rho = 0.95$ (full-sample average)

Hedge Ratio Comparison

- DCC: $h_{\text{DCC}}^* = 0.88 \times \frac{3.5}{4.0} = 0.77$ CCC: $h_{\text{CCC}}^* = 0.95 \times \frac{3.5}{4.0} = 0.83$ OLS: $h_{\text{OLS}}^* = 0.91$

Consequence of Over-hedging

- CCC over-hedges by **6 p.p.** ($0.83 - 0.77$); OLS by **14 p.p.** ($0.91 - 0.77$)
- Over-hedging $\Rightarrow$ excess short futures $\Rightarrow$ if basis widens (common in GFC), both legs lose
- For a \$100M position, 14 p.p. over-hedging = \$14M excess short futures exposure

Case Study 3: HE Decomposition — Variance vs. Correlation

Decomposing Hedging Effectiveness

- The hedge ratio $h_t^* = \rho_t \cdot \frac{\sigma_{\text{spot},t}}{\sigma_{\text{fut},t}}$ has two dynamic components:
- **Variance ratio:** $\sigma_{\text{spot},t}/\sigma_{\text{fut},t}$ — changes when vol regimes differ across markets
- **Correlation:** ρ_t — changes when market microstructure shifts
- CCC captures only the first; DCC captures both

Contribution to h_t^* Variation (S&P 500 – E-mini, 2004–2024)

Source	Calm periods	Crisis periods
Variance ratio variation	75%	45%
Correlation variation	25%	55%

Implication

- In calm periods, CCC captures 75% of hedge ratio dynamics $\Rightarrow$ nearly as good as DCC
- In crises, correlation variation becomes **dominant** — CCC misses 55% of the signal

Case Study 3: Practical Considerations

Implementation Challenges

- ▣ **Margin requirements:** dynamic hedging changes the number of futures contracts daily $\Rightarrow$ margin account fluctuates $\Rightarrow$ cash management needed
- ▣ **Discrete rebalancing:** cannot trade fractional contracts — rounding errors accumulate
- ▣ **Estimation risk:** DCC parameters are estimated with uncertainty; misspecified α, β produce noisy ρ_t
- ▣ **Roll costs:** E-mini contracts expire quarterly — rolling introduces 2–5 bps cost per roll

Mitigation Strategies

- ▣ Rebalance only when $|h_t^* - h_{t-1}^*| > \delta$ (“deadband” strategy) — reduces turnover by 40%
- ▣ Use rolling estimation window (e.g., 1000 days) to keep DCC parameters fresh
- ▣ Complement with position limits: cap $h_t^* \in [0.7, 1.1]$ to prevent extreme ratios from estimation errors

Case Study 3: Cross-Hedging — When Exact Futures Don't Exist

Cross-Hedging with DCC

- Not all assets have liquid futures (e.g., individual stocks, emerging markets)
- Cross-hedge:** use a correlated, liquid proxy (e.g., hedge DAX stock with Euro Stoxx 50 futures)
- DCC hedge ratio: $h_t^* = \rho_t^{\text{cross}} \cdot \frac{\sigma_{\text{asset},t}}{\sigma_{\text{proxy},t}}$
- Key challenge: ρ_t^{cross} is lower and more volatile than same-asset ρ_t

Example: Hedging BET (Romania) with Euro Stoxx 50

	ρ_t	h_t^*	HE
Calm periods	0.52	0.48	68%
Crisis periods	0.71	0.65	82%

- Cross-hedging is **more** effective during crises (correlations converge) — exactly when hedging matters most.

Case Study 3: Decision Framework — Choosing the Right Hedge Model

Model Selection Guide

Criterion	OLS	CCC	DCC
Computational cost	Low	Medium	High
Data requirement	60+ obs	500+ obs	1000+ obs
Correlation regime	Stable	Stable	Time-varying
Best for	FX, rates	Commodities	Equities, HY
Crisis performance	Poor	Medium	Best

Rules of Thumb

- ▣ If $\text{Var}(\rho_t) < 0.01$ in your sample $\Rightarrow$ CCC is sufficient
- ▣ If max drawdown control is the priority $\Rightarrow$ DCC
- ▣ If rebalancing is expensive (> 20 bps) $\Rightarrow$ OLS with monthly review
- ▣ **Always:** backtest the hedge on out-of-sample data before implementation

Case Study 4: Portfolio VaR with Multivariate GARCH

Learning goal: compute portfolio VaR from DCC, backtest it, understand why static VaR fails Basel.

Portfolio VaR and ES Formulas

- **Portfolio volatility:** $\sigma_{p,t} = \sqrt{\mathbf{w}'\mathbf{H}_t\mathbf{w}}$
- **VaR:** $\text{VaR}_t^\alpha = z_\alpha \cdot \sigma_{p,t}$ **ES:** $\text{ES}_t^\alpha = \frac{\phi(z_\alpha)}{\alpha} \cdot \sigma_{p,t}$

Backtesting: Christoffersen Test (1998)

- **Violation:** $I_t = 1\{r_{p,t} < -\text{VaR}_t^\alpha\}$ — should be i.i.d. Bernoulli(α)
- **Unconditional coverage (UC) test:** is the violation proportion = α ?
- **Independence test:** violations must not be clustered

Case Study 4: Why Multivariate VaR?

Portfolio VaR $\neq$ Sum of Individual VaRs

- ▣ **Naive sum:** $\text{VaR}_{\text{sum}} = \sum_i w_i \cdot \text{VaR}_i$ — assumes $\rho = 1$ (worst case)
- ▣ **Portfolio VaR:** $\text{VaR}_p = z_\alpha \cdot \sqrt{w'H_t w}$ — accounts for diversification

Example: 60/40 S&P/DAX, $\rho_t = 0.73$

- ▣ $\text{VaR}_{\text{sum}} = 4.47\%$ vs. $\text{VaR}_{\text{portfolio}} = 4.16\%$ — **7% reduction** from diversification
- ▣ If $\rho_t \rightarrow 0.95$ (crisis): benefit drops to only 2%

Key Insight

- ▣ Diversification benefit is **largest** when ρ is low and **vanishes** during crises — static VaR misses this.

Case Study 4: Numerical Example — Portfolio VaR with DCC

Portfolio: 60% S&P 500, 40% DAX — day t

- Conditional volatilities: $\sigma_{1,t} = 1.8\%$, $\sigma_{2,t} = 2.1\%$; DCC correlation: $\rho_{12,t} = 0.73$
- Covariance matrix: $H_t = \begin{pmatrix} 3.24 & 2.76 \\ 2.76 & 4.41 \end{pmatrix} \times 10^{-4}$

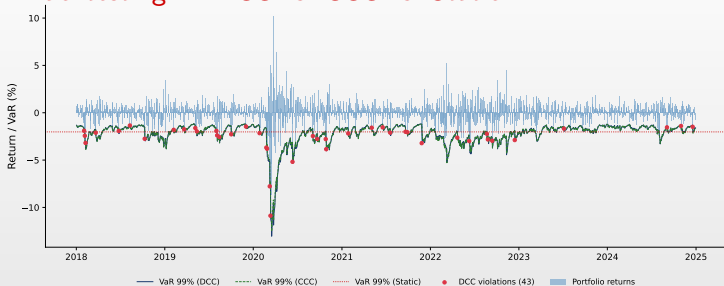
Computing 99% VaR and ES

- $w = (0.6, 0.4)'$, $z_{0.01} = -2.326$
- $\sigma_{p,t}^2 = 0.36 \cdot 3.24 + 0.16 \cdot 4.41 + 2 \cdot 0.24 \cdot 2.76 = 3.197 \times 10^{-4} \Rightarrow \sigma_{p,t} = 1.788\%$
- $\text{VaR}_{99\%} = 2.326 \times 1.788\% = 4.16\%$
- $\text{ES}_{99\%} = \frac{\phi(-2.326)}{0.01} \times 1.788\% = 4.79\%$

Comparison: if ρ is constant ($= 0.5$)

- $\text{VaR} = 3.72\%$, $\text{ES} = 4.28\%$ — underestimates risk by **0.44 p.p. (VaR)** and **0.51 p.p. (ES)**

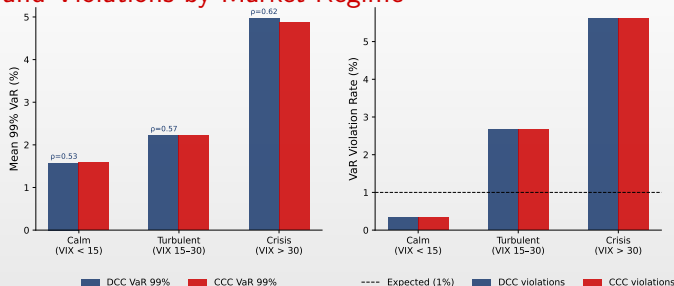
Case Study 4: VaR Backtesting — DCC vs. CCC vs. Static



Interpretation

- **DCC** VaR adjusts quickly to rising crisis correlations — fewer violations
- **CCC/static** VaR underestimates risk in turbulent periods (correlations assumed constant)
- Clustered violations signal failure of the Christoffersen independence test

Case Study 4: VaR and Violations by Market Regime



Conclusions

- DCC VaR rises **automatically** with market regime: ρ_t increases with VIX
- CCC underestimates crisis VaR (VIX > 30) by up to **30%** — excess violations
- **Lesson:** choose DCC/aDCC for assets that become more correlated during crises (equities, HY bonds)

Case Study 4: Christoffersen Test Results

Backtesting 99% VaR (60/40 S&P 500/DAX, 2004–2024, $T = 5,000$)

Model	Count	Violations		Christoffersen p -values		
		Rate	Expected	UC	Ind	CC
DCC	52	1.04%	1%	0.82	0.45	0.67
CCC	71	1.42%	1%	0.02	0.03	0.01
Static	89	1.78%	1%	<0.01	<0.01	<0.01

Interpretation of Test Results

- **DCC**: passes all three tests — violation rate close to nominal 1%, no clustering
- **CCC**: fails UC ($p = 0.02$) and independence ($p = 0.03$) — too many violations, and they cluster during crises
- **Static**: fails all tests decisively — constant covariance is inadequate for 20-year horizons
- Basel III requires $p > 0.05$ for internal VaR model approval — **only DCC qualifies**

Case Study 4: The Kupiec Test — Unconditional Coverage

Setup (Kupiec, 1995)

- ▣ Violation indicator: $I_t = 1\{r_{p,t} < -\text{VaR}_t^\alpha\}$, violations: $x = \sum_{t=1}^T I_t$, rate: $\hat{\pi} = x/T$
- ▣ H_0 : $\hat{\pi} = \alpha$ (model produces the correct coverage)

Likelihood Ratio Statistic

- ▣ Under H_0 : $x \sim \text{Binomial}(T, \alpha)$. The LR test statistic:

$$LR_{UC} = -2 \ln \frac{\alpha^x (1 - \alpha)^{T-x}}{\hat{\pi}^x (1 - \hat{\pi})^{T-x}} \xrightarrow{d} \chi^2(1) \quad \text{under } H_0$$

- ▣ Reject H_0 if $LR_{UC} > \chi_{1,0.05}^2 = 3.84$

Kupiec vs. Christoffersen

- ▣ **Kupiec** tests only **coverage** (“are there too many violations?”)
- ▣ **Christoffersen** adds **independence** test (“are violations clustered?”) — more powerful
- ▣ **Combined** (CC) = UC + Independence: the recommended approach

Case Study 4: Basel Traffic Light System

Regulatory VaR Model Assessment (Basel Committee, 250-day window)

Zone	Violations	Capital multiplier	Action
Green	0–4	3.0×	Model accepted
Yellow	5–9	3.4–3.85×	Supervisory review
Red	10+	4.0×	Model rejected

How Do Our Models Score? (per 250-day window)

Model	Avg. violations	Max. violations	Zone
DCC	2.6	4	Green
CCC	3.6	7	Yellow (in crisis years)
Static	4.5	11	Red (in crisis years)

Implication

- ☐ Only DCC stays in the **green zone** consistently — banks using CCC/static face **higher capital charges** during crises

Case Study 4: ES vs. VaR — Why Regulators Prefer ES

Limitations of VaR

- ▣ VaR answers: *“What is the maximum loss with 99% confidence?”*
- ▣ Says **nothing** about loss size in the 1% tail
- ▣ **Not subadditive**: $\text{VaR}(A + B)$ can exceed $\text{VaR}(A) + \text{VaR}(B)$

Expected Shortfall (ES) — Coherent Risk Measure

- ▣ ES answers: *“Given that we’re in the 1% tail, how bad is the average loss?”*
- ▣ $\text{ES}_\alpha = \mathbb{E}[L | L > \text{VaR}_\alpha]$ — always $\geq \text{VaR}$
- ▣ **Subadditive**: $\text{ES}(A + B) \leq \text{ES}(A) + \text{ES}(B)$ — diversification always recognized

FRTB Transition (2023+)

- ▣ **FRTB** (*Fundamental Review of the Trading Book*) replaces 99% VaR with 97.5% ES as the primary capital metric
- ▣ Multivariate GARCH provides the H_t needed for both VaR and ES computation
- ▣ DCC-ES captures **tail dependence** that DCC-VaR alone misses

Case Study 4: ES Backtesting — Acerbi & Szekely (2014)

The ES Backtesting Problem

- VaR: counts violations (binary). ES: evaluates **how large** losses are when VaR is exceeded — harder
- ES is **not elicitable** (Gneiting, 2011) — no simple scoring rule exists for ES alone

Acerbi–Szekely Test Statistic

- H_0 : ES model is correctly specified. Define: $Z_t = \frac{r_{p,t}}{\text{ES}_t^\alpha} \cdot 1\{r_{p,t} < -\text{VaR}_t^\alpha\} + 1$
- Under H_0 : $\mathbb{E}[Z_t] = 0$. Test statistic: $\bar{Z} = \frac{1}{T} \sum_{t=1}^T Z_t$
- $\bar{Z} < 0$: ES **underestimates** tail losses; $\bar{Z} > 0$: ES conservative. p -value via simulation

Practical Implications

- Under FRTB, banks must backtest **both** VaR (Christoffersen/Kupiec) and ES (Acerbi–Szekely)
- DCC with Student- t innovations passes ES backtests; DCC Gaussian often fails

Case Study 4: Multi-Horizon VaR — Scaling and Aggregation

Scaling VaR Across Horizons

- ▣ **Square-root-of-time rule:** $\text{VaR}_h = \text{VaR}_1 \times \sqrt{h}$ — assumes i.i.d. returns
- ▣ With GARCH: rule **overestimates** long-horizon VaR (ignores mean-reversion in σ_t)
- ▣ DCC approach: simulate multi-step $H_{t+1}, \dots, H_{t+h}$ via GARCH recursion

Example: 1-day vs. 10-day VaR (60/40 portfolio)

	1-day VaR	10-day VaR	Scaling
$\sqrt{T}$ rule	4.16%	13.16%	$\sqrt{10} = 3.16$
DCC simulation	4.16%	11.84%	2.85

- ▣ $\sqrt{T}$ rule overestimates 10-day VaR by **11%** (ignores mean-reversion in σ_t)

Basel Requirement

- ▣ Basel requires 10-day VaR for market risk capital
- ▣ Using DCC simulation instead of $\sqrt{T}$ scaling can reduce capital charges by 8–15%

Case Study 4: Summary — Key Numbers

Reference Table: VaR Backtesting Results (2004–2024)

Statistic	Value
DCC violation rate (nominal 1%)	1.04%
CCC violation rate	1.42% (+42% excess)
Static violation rate	1.78% (+78% excess)
DCC: Christoffersen CC p -value	0.67 (pass)
CCC: Christoffersen CC p -value	0.01 (fail)
CCC crisis VaR underestimation	up to 30%
Basel zone (DCC)	Green always
Basel zone (Static, crisis)	Red
Diversification benefit ($\rho = 0.5$)	15% VaR reduction
Diversification benefit ($\rho = 0.95$)	2% VaR reduction

Bottom Line

- DCC is the only model that consistently passes regulatory backtesting across all market regimes
- The cost of using CCC or static models is both regulatory (higher capital) and economic (tail risk underestimation)

Case Study 5: What is Systemic Risk?

Learning goal: connect DCC correlations to systemic risk measures (ΔCoVaR , MES , SRISK) and G-SIB identification.

Definition

- ▣ **Systemic risk:** distress of one institution triggers cascading failures throughout the financial system

Key Characteristics

- ▣ **Too Big / Too Connected to Fail:** failure destabilizes the system via counterparty contagion
- ▣ **Correlation surge:** in crises, correlations spike $\Rightarrow$ diversification vanishes
- ▣ **Non-linearity:** systemic risk emerges from **interactions**, not from summing individual risks

Why MGARCH?

- ▣ Systemic risk = **joint tail behavior** of multiple institutions
- ▣ DCC-GARCH provides the time-varying H_t needed to compute conditional risk measures
- ▣ Static correlations miss the crisis amplification that defines systemic events

Case Study 5: Systemic Risk Monitoring

MGARCH-Based Systemic Risk Indicators

- ▣ **CoVaR** (*Conditional Value at Risk*) (Adrian & Brunnermeier, 2016): system VaR conditional on one institution's stress

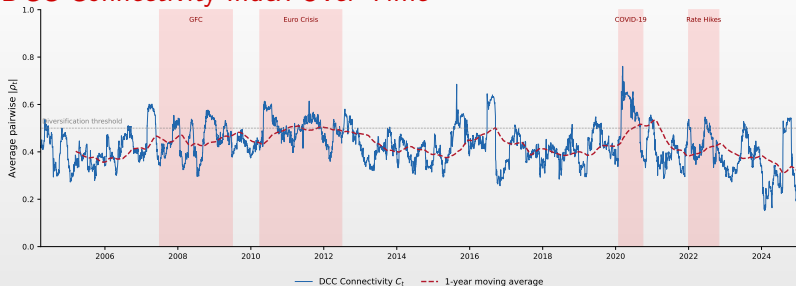
$$\Delta \text{CoVaR}_{i,t} = \text{VaR}_t^{\text{system}|i \text{ stressed}} - \text{VaR}_t^{\text{system}|i \text{ normal}}$$

- ▣ **DCC connectivity index**: $C_t = \frac{2}{N(N-1)} \sum_{i < j} |\rho_{ij,t}|$ — average correlation
- ▣ When $C_t \rightarrow 1$ simultaneously $\Rightarrow$ diversification vanishes $\Rightarrow$ **crisis signal**

Regulatory Requirements

- ▣ **Basel III/IV**: banks must report VaR at multiple horizons with dynamic correlations
- ▣ **FRTB** (*Fundamental Review of the Trading Book*): replaces VaR with ES, requires multivariate GARCH
- ▣ **DCC-GARCH** provides the H_t matrix needed for CoVaR and stress testing

Case Study 5: DCC Connectivity Index Over Time



Interpretation

- C_t jumps from ~ 0.45 (normal) to > 0.85 during GFC and COVID — **diversification collapses**
- The spike in connectivity **precedes** peak drawdowns by 2–3 weeks — early warning signal
- Post-crisis normalization takes 6–12 months, creating a window of **false security**
- When $C_t > 0.5$: portfolio benefits of diversification are **significantly reduced**

Case Study 5: Numerical Example — ΔCoVaR

4-bank system: $\sigma_i = 2.5\%$, $\rho_{i,\text{sys}} = 0.72$, $\sigma_{\text{sys}} = 1.8\%$

- Normal: $\text{VaR}^{\text{sys}} = 2.326 \times 1.8\% = 4.19\%$. Under stress ($\kappa \approx 2.5$):

$$\sigma_{\text{sys}|i} = \sigma_{\text{sys}} \sqrt{1 + \rho_{i,\text{sys}}^2 (\kappa - 1)} = 1.8\% \times 1.33 = 2.39\%$$

Result

- $\text{CoVaR}^{\text{sys}|i} = 2.326 \times 2.39\% = 5.56\%$
- $\Delta\text{CoVaR}_i = 5.56\% - 4.19\% = 1.37\%$ — bank i 's stress adds **33%** to system risk

Policy Implication

- Banks with high ΔCoVaR face **higher capital surcharges** (G-SIB (*Global Systemically Important Bank*) buffers)
- DCC is essential: with CCC (ρ fixed), ΔCoVaR misses crisis amplification
- During GFC, ΔCoVaR for top-5 US banks exceeded **3%** — three times normal

Case Study 5: MES — Marginal Expected Shortfall

Definition (Acharya et al., 2017)

- **MES**: expected loss of institution i when the system is in the tail:

$$\text{MES}_{i,t} = \mathbb{E} [r_{i,t} \mid r_{\text{sys},t} < \text{VaR}_{\text{sys},t}^{\alpha}]$$

- DCC: $\mathbb{E}[r_i | r_{\text{sys}}] = \rho_{i,\text{sys},t} \cdot \frac{\sigma_{i,t}}{\sigma_{\text{sys},t}} \cdot r_{\text{sys}}$

Numerical Example

- Bank i : $\sigma_i = 2.5\%$, $\rho_{i,\text{sys}} = 0.72$; System: $\sigma_{\text{sys}} = 1.8\%$
- On a 2σ system crash ($r_{\text{sys}} = -3.6\%$): $\text{MES}_i = 0.72 \times \frac{2.5}{1.8} \times 3.6\% = 3.60\%$

MES vs. CoVaR

- MES measures the contribution of **bank to system** (exposure)
- CoVaR measures the impact of **system on bank** (contagion)
- Both require DCC-GARCH

Case Study 5: SRISK — Capital Shortfall Under Stress

Definition (Brownlees & Engle, 2017)

- ▣ **SRISK**: expected capital shortfall during a systemic crisis

$$\text{SRISK}_{i,t} = k \cdot D_{i,t} - (1 - k) \cdot W_{i,t} \cdot (1 - \text{LRMES}_{i,t})$$

- ▣ $k = 8\%$ (prudential ratio), D : debt, W : market cap, **LRMES** (*Long-Run Marginal Expected Shortfall*): equity loss in a 40% market decline (6 months)

Role of DCC-GARCH in SRISK

- ▣ DCC provides $\rho_{i,\text{sys},t}$ and $\sigma_{i,t} \Rightarrow \text{MES} \Rightarrow \text{LRMES}$ (6-month DCC simulation)
- ▣ SRISK is updated **daily** as market cap and DCC correlations change

Application

- ▣ NYU Stern's V-Lab publishes **daily SRISK rankings** for 1200+ global firms, all based on DCC-GARCH: `vlab.stern.nyu.edu`

Case Study 5: G-SIB Identification and Capital Surcharges

How Regulators Identify Systemically Important Banks

- ▣ **FSB/Basel:** G-SIB score = size + interconnectedness + substitutability + complexity + cross-border
- ▣ **Market-based:** ΔCoVaR , MES, SRISK complement regulatory scores

G-SIB Capital Surcharges (2024)

Bucket	Surcharge	Example banks
5 (highest)	3.5%	JPMorgan
4	2.5%	HSBC, Citi
3	2.0%	BNP Paribas, Goldman Sachs
2	1.5%	Deutsche Bank, UBS
1	1.0%	Santander, Standard Chartered

Key Point

- ▣ Higher DCC connectivity $\Rightarrow$ higher systemic importance $\Rightarrow$ higher capital buffers $\Rightarrow$ lower ROE
- ▣ Measuring $\rho_{ij,t}$ accurately has direct **financial consequences** for banks

Case Study 5: Contagion Networks — Visualizing Interconnectedness

From Pairwise Correlations to Network Topology

- DCC produces $N(N - 1)/2$ pairwise $\rho_{ij,t} \Rightarrow$ **weighted graph** (nodes = institutions, edges = $|\rho_{ij,t}|$)
- Metrics: **degree centrality**, **betweenness** (bridging role), **eigenvector centrality**

Crisis vs. Normal Network Structure

Network property	Normal	Crisis
Average $ \rho_{ij} $	0.45	0.85
Network density	0.35	0.90
Clustering coefficient	0.52	0.88
Most central node	JPMorgan	JPMorgan

Insight

- In normal times, the network is **sparse** — shocks stay local
- In crises, it becomes **dense and fully connected** — every node affects every other, making diversification impossible

Case Study 5: DCC for Regulatory Stress Testing

How Central Banks Use DCC-GARCH

- ▣ **ECB (*European Central Bank*)/Fed stress tests:** simulate asset returns under adverse scenarios
- ▣ DCC provides the **correlation structure**: “if equities fall 40%, what happens to credit spreads?”
- ▣ Key: use **crisis-calibrated** DCC parameters, not full-sample averages

Stress Testing Workflow

1. Fit DCC-GARCH to 10+ years of data
2. Extract crisis-period α, β parameters (e.g., GFC subsample)
3. Simulate 10,000 paths of $H_{t+1}, \dots, H_{t+h}$ under stressed parameters
4. Compute portfolio loss distribution $\Rightarrow$ stressed VaR, ES, SRISK

Why Not Historical Simulation?

- ▣ Historical simulation uses **past** returns directly — but no two crises are alike
- ▣ DCC stress testing generates **new** crisis scenarios consistent with the fitted correlation dynamics

Case Study 5: Summary — Systemic Risk Toolkit

DCC-Based Systemic Risk Measures

Measure	What it captures	Key input from DCC
C_t (connectivity)	System-wide correlation	$ \rho_{ij,t} $ average
ΔCoVaR	Impact of one bank on system VaR	$\rho_{i,\text{sys},t}, \sigma_{i,t}$
MES	Bank loss during system crash	$\rho_{i,\text{sys},t}, \sigma_{i,t}$
SRISK	Capital shortfall under stress	LRMES from DCC simulation
DY spillover	Cross-market volatility transmission	VAR on DCC residuals

Key Takeaways

- DCC-GARCH is the **backbone** of modern systemic risk measurement
- All major indicators (ΔCoVaR , MES, SRISK) require time-varying $\rho_{ij,t}$
- Connectivity index $C_t > 0.7$ is a reliable **early warning signal** for systemic crises
- Regulatory applications: G-SIB identification, capital surcharges, stress testing

Realized Covariance and HAR-DRD

From Parametric to Nonparametric: Realized Covariance

- ▣ Instead of *modeling* H_t , **measure** it directly from intraday data: $RC_t = \sum_{k=1}^M r_{t,k} r'_{t,k}$ (M = intraday intervals, e.g., 5-min)
- ▣ **Model-free**: no GARCH assumptions needed — the data “speak for themselves”
- ▣ Requires **high-frequency data** — not available for all assets (bonds, OTC derivatives)

HAR-DRD Model (Shephard & Sheppard, 2010; Oh & Patton, 2016)

- ▣ Decompose: $RC_t = D_t^{RC} R_t^{RC} D_t^{RC}$ (analogous to DCC but using realized measures)
- ▣ Realized volatilities with HAR dynamics: $d_{i,t+1} = c + \beta_d d_{i,t} + \beta_w \bar{d}_{i,t}^{(w)} + \beta_m \bar{d}_{i,t}^{(m)} + \varepsilon_{t+1}$
- ▣ $\bar{d}^{(w)}$, $\bar{d}^{(m)}$ = weekly, monthly averages — captures heterogeneous traders

When to Use

- ▣ **Preferred** when high-frequency data is available — dominates MGARCH in forecasting (Lunde et al., 2016)

DCC-MIDAS: Separating Short-Run and Long-Run Dynamics

Motivation

- Standard DCC has **one time scale** — cannot distinguish daily fluctuations from secular trends
- Correlations have **long-run component** (macro factors) and **short-run component** (news)

DCC-MIDAS (Colacito, Engle & Ghysels, 2011)

- Time-varying $\bar{Q}_t$: $\bar{q}_{ij,t} = \sum_{k=1}^K \varphi_k(\omega_1, \omega_2) c_{ij,t-k}$, $c_{ij,t} = \frac{1}{N_v} \sum_{n=1}^{N_v} z_{i,t_n} z_{j,t_n}$
- φ_k = Beta-weighting scheme (MIDAS — *Mixed Data Sampling*)
- Short-run DCC dynamics around the slowly moving $\bar{q}_{ij,t}$

Advantage over Standard DCC

- Captures **structural breaks** naturally — $\bar{q}_{ij,t}$ adapts to new correlation regimes
- Better long-horizon forecasts (1 month+) — standard DCC reverts too quickly to $\bar{Q}$

Machine Learning for Covariance Forecasting

Why ML? Limitations of Parametric MGARCH

- ▣ MGARCH assumes a **fixed functional form** — ML can learn nonlinear dynamics directly from data
- ▣ High-dimensional forecasting ($N > 100$) intractable for MGARCH — ML scales better

Approaches (Bucci, 2020; De Nard et al., 2021; Zhang et al., 2023)

- ▣ **LSTM / GRU**: treat $\text{vech}(H_t)$ as multivariate time series — capture long-range dependencies
- ▣ **Transformers**: attention identifies which past correlations are most relevant
- ▣ **Graph Neural Networks**: model asset relationships as graph — natural for contagion analysis
- ▣ **Hybrid DCC-ML**: DCC as baseline, ML corrects residuals (Gu, Kelly & Xiu, 2020)

Current Status

- ▣ ML shows promise for **large N** and **nonlinear regimes**, but lacks interpretability
- ▣ **No consensus** that ML consistently beats DCC for moderate N (< 50)
- ▣ Hybrid approaches (DCC + ML corrections) are the most promising direction

Conformal Prediction for Covariance Calibration

The Calibration Problem

- ▣ MGARCH VaR assumes a **specific distribution** (Gaussian, Student- t) — misspecification leads to poor coverage
- ▣ Even with correct covariance, the **distributional assumption** can fail in the tails

Conformal Prediction (Vovk et al., 2005; Gibbs & Candès, 2021)

- ▣ **Distribution-free** coverage guarantee: prediction sets with exact $1 - \alpha$ coverage
- ▣ Applied to VaR: adjust quantile using past residuals, not parametric assumptions
- ▣ **Adaptive Conformal Inference (ACI)**: tightens in calm, widens in crises

Integration with MGARCH

- ▣ DCC-GARCH for **point forecast** of H_t , then conformalize VaR/ES bounds
- ▣ **Guaranteed coverage** even when DCC is misspecified — best of both worlds

Summary: Key Takeaways

What We Learned

1. Multivariate GARCH models the **time-varying covariance matrix** H_t
2. Main challenge: **curse of dimensionality** — $N(N + 1)/2$ unique elements
3. **VEC/BEKK**: direct parameterization — flexible but limited to small N
4. **CCC**: constant correlations — parsimonious but unrealistic
5. **DCC**: dynamic correlations with only 2 extra parameters — the workhorse model
6. **aDCC**: adds asymmetry in correlations (leverage effect)
7. **Factor/GO-GARCH**: for large N via dimensionality reduction
8. Two-step estimation makes DCC/CCC feasible for hundreds of assets

Central Message

- ☐ Correlations are **not constant**. Ignoring time-varying correlations leads to **underestimation of portfolio risk** and **suboptimal hedging decisions**.

Practical Pipeline: Data Preparation

Step 1: Data Collection and Cleaning

- ▣ Use **log-returns** $\times 100$ (percentage) — stabilizes optimization
- ▣ Align all series on **common trading dates** (remove holidays, non-overlapping days)
- ▣ Handle **nonsynchronous trading**: open-to-open returns or Scholes–Williams correction

Step 2: Preliminary Analysis

- ▣ **Rolling correlations** (60-day, 252-day) — visual check for time variation
- ▣ Test **constant correlation**: Engle–Sheppard (2001) or Tse (2000) LM
- ▣ Test **ARCH effects**: Engle (1982) LM on squared returns
- ▣ Check **structural breaks**: correlations vs. VIX

Decision Point

- ▣ CCC **not rejected**: use CCC (simpler, fewer parameters)
- ▣ CCC **rejected**: proceed to DCC/aDCC

Practical Pipeline: Estimation and Model Selection

Step 3: Univariate GARCH Specification

- ▣ Fit **GARCH(1,1)** or **GJR-GARCH(1,1)** to each series individually
- ▣ Use AIC/BIC to select the best univariate specification **per asset**
- ▣ Check: $\alpha + \beta < 1$ (stationarity), residuals pass Ljung–Box test

Step 4: Multivariate Model Estimation

- ▣ $N \leq 5$: estimate DCC, aDCC, Diagonal BEKK; compare with AIC/BIC
- ▣ $5 < N \leq 50$: DCC and aDCC only (BEKK infeasible)
- ▣ $N > 50$: DCC with composite likelihood, or Factor GARCH
- ▣ Initialize from **EWMA** ($\lambda = 0.94$) or sample correlation matrix

Step 5: Diagnostics

- ▣ Standardized residuals $\hat{z}_{it}$: no serial correlation (Ljung–Box), no ARCH effects
- ▣ Cross-products $\hat{z}_{it}\hat{z}_{jt}$: no cross-correlation (Hosking, 1980)
- ▣ CCC vs. DCC: $LR = 2(\mathcal{L}_{DCC} - \mathcal{L}_{CCC}) \sim \chi^2(2)$

Practical Pipeline: Validation and Application

Step 6: Out-of-Sample Validation

- ▣ Data: **70% training, 30% testing** (or expanding window)
- ▣ Forecast $\hat{H}_{t+1}$ one step ahead, compute portfolio VaR/ES
- ▣ Backtest: Kupiec (UC), Christoffersen (CC), Acerbi–Szekely (ES)
- ▣ Compare models: **Model Confidence Sets** or Diebold–Mariano on loss functions

Step 7: Application

- ▣ **Portfolio optimization:** $w_t = H_t^{-1}1/(1'H_t^{-1}1)$ (minimum variance)
- ▣ **Hedging:** $h_t^* = \sigma_{12,t}/\sigma_{22,t}$ (dynamic hedge ratio)
- ▣ **Risk reporting:** daily VaR/ES with time-varying correlations
- ▣ **Re-estimate** every 1–2 years or after major structural breaks

Golden Rule

- ▣ The **simplest model that passes out-of-sample backtests** is the best model

Python: DCC-GARCH Estimation and Portfolio VaR

```
import numpy as np
from arch import arch_model
from scipy.optimize import minimize

# Step 1: Fit univariate GARCH(1,1) to each series
models, std_resids = [], []
for col in returns.columns:
    mod = arch_model(returns[col]*100, vol='Garch', p=1, q=1, dist='t')
    res = mod.fit(disp='off')
    models.append(res); std_resids.append(res.std_resid)
Z = np.column_stack(std_resids) # T x N standardized residuals
Q_bar = np.corrcoef(Z.T) # unconditional correlation

# Step 2: DCC dynamics (a, b) via composite likelihood
def dcc_loglik(params, Z, Q_bar):
    a, b = params; T, N = Z.shape; Qt = Q_bar.copy(); ll = 0.0
    for t in range(T):
        zt = Z[t:t+1].T
        Qt = (1-a-b)*Q_bar + a*(zt @ zt.T) + b*Qt
        Dt_inv = np.diag(1.0 / np.sqrt(np.diag(Qt)))
        Rt = Dt_inv @ Qt @ Dt_inv
        ll += np.log(np.linalg.det(Rt)) + zt.T @ np.linalg.inv(Rt) @ zt - zt.T @ zt
    return 0.5 * float(ll)
res = minimize(dcc_loglik, [0.05, 0.90], args=(Z, Q_bar),
               bounds=[(1e-6,0.5),(1e-6,0.9999)], method='L-BFGS-B')

# Step 3: Portfolio VaR (99%)
w = np.array([0.6, 0.4]) # portfolio weights
port_var = np.sqrt(w @ H_t @ w) * 2.326 # 99% z-score
```

Model Selection Decision Tree

Step 1: How many assets (N)?

- ▣ $N \leq 3$: all models (BEKK, CCC, DCC, aDCC)
- ▣ $3 < N \leq 50$: DCC / aDCC / GO-GARCH
- ▣ $N > 50$: DCC, Factor GARCH, shrinkage

Step 2: Do you need volatility spillovers?

- ▣ **Yes** (contagion, systemic risk): BEKK ($N \leq 5$) or Diebold–Yilmaz
- ▣ **No** (VaR, hedging): → Step 3

Step 3: Are correlations time-varying?

- ▣ Run Engle–Sheppard or Tse LM test
- ▣ Not rejected → CCC
- ▣ Rejected → DCC, then: asymmetry?

Step 4: Asymmetric correlations?

- ▣ Test $g = 0$ in aDCC: significant → aDCC; otherwise → DCC
- ▣ **Final**: compare via out-of-sample backtest (not AIC)

Key Formulas: Quick Reference Card

Model	Core Equation	Parameters
VEC	$\text{vech}(H_t) = c + A \text{vech}(\varepsilon_{t-1} \varepsilon'_{t-1}) + B \text{vech}(H_{t-1})$	$N^*(1 + 2N^*)$
BEKK	$H_t = CC' + A' \varepsilon_{t-1} \varepsilon'_{t-1} A + B' H_{t-1} B$	$N(N+1)/2 + 2N^2$
CCC	$H_t = D_t R D_t, \quad h_{ii,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i h_{ii,t-1}$	$3N + N(N-1)/2$
DCC	$Q_t = (1-a-b)\bar{Q} + a z_{t-1} z'_{t-1} + b Q_{t-1}$	$3N + N(N-1)/2 + 2$
aDCC	$Q_t = \bar{Q}^* + a \tilde{z}_{t-1} \tilde{z}'_{t-1} + g n_{t-1} n'_{t-1} + b Q_{t-1}$	$3N + N(N-1)/2 + 3$

Key Quantity	Formula
Portfolio variance	$\sigma_{p,t}^2 = w' H_t w$
Min-variance hedge	$h_t^* = \sigma_{12,t} / \sigma_{22,t}$
Portfolio VaR (99%)	$\text{VaR}_{0.99} = -\mu_{p,t} + z_{0.99} \sqrt{w' H_t w}$
DCC stationarity	$a + b < 1$
BEKK stationarity	$\rho(A \otimes A + B \otimes B) < 1$

Quiz: Multivariate GARCH (1/2)

Question 1

- Why does CCC assume $H_t = D_t R D_t$ instead of directly modeling all elements of H_t ?

Question 2

- In the DCC model, what happens to the dynamic correlations when $a = b = 0$?

Question 3

- You have a portfolio of 200 stocks. Which multivariate GARCH model would you recommend and why?

Question 4

- Why do correlations increase during financial crises? How does DCC capture this?

Quiz: Multivariate GARCH (2/2)

Question 5

- What is the main advantage of BEKK over DCC? And its main disadvantage?

Question 6

- A hedge fund has a portfolio of 50 stocks and wants to compute daily VaR. Which multivariate GARCH model is most appropriate? What trade-offs does it involve?

Question 7

- What is nonsynchronous trading and how does it affect the estimation of correlations between markets in different time zones?

Question 8

- How does aDCC differ from standard DCC? In which practical situations would aDCC perform significantly better?

Quiz: Multivariate GARCH (3/3) — Exam-Level

Question 9 (Analytical)

- Show that DCC rescaling $R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$ guarantees $R_{ii,t} = 1 \forall i$. When is R_t p.d.?

Question 10 (Applied)

- DCC-GARCH, 60/40 equity–bond portfolio. Correlation jumps from +0.20 to −0.30 in crisis. Explain the mechanism and impact on portfolio VaR vs. CCC.

Question 11 (Model Selection)

- CCC, DCC, aDCC for 20 stocks: in-sample aDCC wins (AIC); out-of-sample CCC has fewest VaR violations. Which model do you deploy and why?

Question 12 (Conceptual)

- Why do BEKK and DCC answer fundamentally different questions? Give one research question best addressed by each.

Quiz: Answers (1/6)

Answer 1

- ▣ $H_t = D_t R D_t$ separates **variance dynamics** from **correlation structure**
- ▣ Each variance modeled with univariate GARCH; correlation from standardized residuals
- ▣ P.d. guaranteed if R is p.d. and all $h_{ii,t} > 0$
- ▣ Direct modeling of all $N(N+1)/2$ elements: too many parameters, p.d. hard to enforce

Answer 2

- ▣ When $a = b = 0$: $Q_t = \bar{Q} \forall t$, so $R_t = \bar{R}$ (constant)
- ▣ DCC **reduces to CCC** — CCC is nested as a special case

Quiz: Answers (2/6)

Answer 3

- ▣ $N = 200$: **Factor GARCH** or **DCC** with composite likelihood
- ▣ Full BEKK infeasible; DCC needs only 2 correlation parameters regardless of N
- ▣ Factor models reduce dimensionality by extracting $K \ll N$ factors

Answer 4

- ▣ In crises: **large joint negative shocks** ($z_{1,t-1}, z_{2,t-1} \ll 0$)
- ▣ $q_{12,t}$ increases because $z_{1,t-1}z_{2,t-1} > 0$ is large $\Rightarrow \rho_{12,t}$ rises
- ▣ aDCC captures this even more explicitly via the asymmetric g term

Quiz: Answers (3/6)

Answer 5

- ▣ **Advantage:** BEKK captures *volatility spillovers* — direct transmission through off-diagonal A, B
- ▣ **Disadvantage:** parameter explosion ($N(N+1)/2 + 2N^2$) — impractical for $N > 5$

Answer 6

- ▣ **DCC:** only $3N + 2 = 152$ parameters (vs. BEKK: 6,275)
- ▣ Estimation: 50 univariate GARCHs + 2-parameter optimization
- ▣ Trade-off: DCC **does not capture spillovers**

Quiz: Answers (4/6)

Answer 7

- ▣ Markets in different time zones do not trade simultaneously
- ▣ Close-to-close correlations **underestimate** the true linkage
- ▣ Solution: Scholes–Williams correction or open-to-open returns

Answer 8

- ▣ aDCC adds $g n_{t-1} n'_{t-1}$ — increases correlation more after joint **negative** shocks
- ▣ Performs better for stock–bond pairs or emerging markets with strong leverage effects

Quiz: Answers (5/6)

Answer 9

- ▣ Element (i, i) : $q_{ii,t}/\sqrt{q_{ii,t} \cdot q_{ii,t}} = q_{ii,t}/q_{ii,t} = 1$ — holds for any $q_{ii,t} > 0$
- ▣ R_t is p.d. $\Leftrightarrow Q_t$ is p.d. (rescaling = congruence transformation)
- ▣ $R_t = PQ_tP$ with $P = \text{diag}(Q_t)^{-1/2}$ positive diagonal — preserves definiteness

Answer 10

- ▣ Sign flip $(+0.20 \rightarrow -0.30)$ = **flight-to-quality**: equities fall, bonds rally
- ▣ CCC: constant positive correlation $\Rightarrow$ expects both to fall together $\Rightarrow$ **overestimates** risk
- ▣ DCC: negative correlation correctly shows bonds *hedge* equities $\Rightarrow$ VaR lower and more accurate
- ▣ CCC may trigger unnecessary de-risking — wrong in a dangerous direction

Quiz: Answers (6/6)

Answer 11

- ▣ Deploy **CCC** — out-of-sample performance is the only criterion for risk management
- ▣ aDCC wins in-sample but CCC wins out-of-sample = textbook **overfitting**
- ▣ Extra aDCC parameters capture patterns that do not generalize
- ▣ Golden Rule: the simplest model that passes backtests is the best model

Answer 12

- ▣ **BEKK**: “*where do shocks originate and how do they propagate?*” — directional spillovers via A, B
 - ▶ Ex.: “Does US banking volatility spill over to European banks?”
- ▣ **DCC**: “*how does co-movement change over time?*” — time-varying correlation, no causality
 - ▶ Ex.: “How correlated were stocks and bonds during COVID-19?”
- ▣ BEKK = *structural* model; DCC = *descriptive* model

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